

COPPERBELT UNIVERSITY

School of Engineering

Department of Mechatronics and Electromechanical Engineering

IMPROVEMENT OF A MOTORIZED GYROSCOPE

(Improved Design for Automated, Sustained Operation)

A Thesis Submitted in Partial Fulfilment of the Requirements
for the Award of the Bachelor of Engineering Degree

Group Members:

Name	Student Number	Programme
Janet Mwaba	22107919	Electromechanical Engineering
Luyando Chiyasa	22111262	Mechatronics Engineering
Chabota Mwenda	22176470	Mechatronics Engineering
Isaac Phiri	22105748	Mechatronics Engineering

Supervisor: Mr. Bennet Siyingwa

Year: 2026

DECLARATION

We declare that this project report is entirely our own work and has not been submitted for any other degree or qualification at any other institution. All references and sources of information have been acknowledged in the text and listed in the References section. Any work done by others has been explicitly attributed.

Name	Signature	Date
Janet Mwaba	_____	_____
Luyando Chiyasa	_____	_____
Chabota Mwenda	_____	_____
Isaac Phiri	_____	_____

Supervisor Approval:

Mr. Bennet Siyingwa: _____ Date: _____

ABSTRACT

This thesis documents the design, analysis, manufacture, and testing of an improved motorized gyroscope capable of sustained, automated operation. The work continues from a prototype built by a previous student team at Copperbelt University (CBU) whose system failed to maintain stable high-speed rotation for extended periods due to flywheel imbalance, lack of automated control, and unreliable breadboard electronics.

The redesigned system employs a precision-machined mild steel flywheel of mass 0.5 kg and radius 0.1079 m, yielding a moment of inertia of 0.002910 kg.m². A 950 KV x2212 brushless DC (BLDC) motor drives the flywheel to the nominal operating speed of 3000 RPM. Stability analysis using gyroscopic dynamics confirmed a stability margin factor of approximately 21 at this speed -- meaning the gyroscopic torque (0.4571 N.m) exceeds the maximum gravitational torque by 21 times.

A systematic engineering review of control strategies identified Proportional-Integral-Derivative (PID) control as the most appropriate approach. PID gains were tuned using the MATLAB Simulink PID Tuner, allowing risk-free optimisation before hardware deployment. Simulation predicted a settling time of 1.8 seconds and less than 10 percent overshoot following a 5-degree disturbance; these predictions were confirmed on hardware.

The monitoring infrastructure uses an ESP32 dual-core microcontroller and an MPU6050 six-axis inertial measurement unit (IMU). Sensor data is transmitted to a Node.js server over a serial USB bridge or Wi-Fi TCP connection. A React-based dashboard displays live charts and stat cards via WebSocket, with configurable safety thresholds and automatic emergency stop. PostgreSQL stores all readings, sessions, and alert events.

A complete five-stage test programme was executed: motor control characterisation, IMU accuracy verification, PID performance evaluation, dashboard latency and emergency-stop timing, and an extended stability run. All tests passed their defined acceptance criteria. The extended run achieved 2 hours of continuous stable operation at a mean speed of 2985 RPM with a variation of +/-15 RPM. Maximum tilt deviation was 2.8 degrees, well within the 5-degree stability limit. No mechanical failures or unplanned stops occurred.

The results demonstrate that reliable sustained automated gyroscope operation is achievable with locally procured components and CBU workshop facilities. The engineering methodology -- systematic literature review, simulation before build, and staged testing -- is directly transferable to industrial electromechanical systems

development.

Keywords: gyroscope, BLDC motor, ESP32, MPU6050, PID control, real-time monitoring, dynamic balancing, mechatronics, extended endurance, WebSocket.

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We extend our sincere gratitude to our project supervisor, Mr. Bennet Siyingwa, for his consistent guidance, technical insight, and constructive feedback at every stage of this project -- from the initial problem definition through to the interpretation of the endurance run results. His industry experience informed many of our practical engineering decisions and saved us from several avoidable mistakes.

We are grateful to the CBU School of Engineering for providing access to the workshop, CNC lathe, balancing rig, oscilloscope, and laboratory instruments without which this project could not have been completed to the standard described in this report. We thank the workshop technicians who assisted directly with the CNC turning operation and the dynamic balancing rig setup, and who gave practical machining advice that significantly improved the quality and accuracy of the finished flywheel.

We acknowledge the previous CBU student team whose design documentation, CAD drawings, and partial prototype gave us a well-defined starting point and saved us significant development time. Their work was carefully reviewed and critically assessed as part of our project initiation, and the experience of identifying and justifying improvements to an existing design was itself a valuable engineering exercise.

We thank the Copperbelt University library staff for assistance in locating technical references and for providing access to online journal databases. We also acknowledge the developers and contributors of the open-source tools and libraries used in this project: the Arduino/ESP32 framework, Node.js, React, PostgreSQL, Drizzle ORM, FPDF2, and MATLAB. All software used was legally licensed or open-source.

Finally, we thank our families for their patience, understanding, and practical support throughout four years of engineering study at CBU. The sacrifice of time and the encouragement they provided was a constant source of motivation throughout this challenging and rewarding project.

DEDICATION

To our families, who have supported us through four years of engineering study at Copperbelt University.

And to every engineering student in Zambia who is determined to build something real with whatever is locally available.

TABLE OF CONTENTS

DECLARATION

ABSTRACT

ACKNOWLEDGEMENTS

DEDICATION

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

CHAPTER 1: INTRODUCTION

- 1.1 Background
- 1.2 Problem Statement
- 1.3 Aim
- 1.4 Motivation
- 1.5 Objectives
- 1.6 Scope and Limitations
- 1.7 Report Structure

CHAPTER 2: LITERATURE REVIEW

- 2.1 Gyroscope Theory and Angular Momentum
- 2.2 Review of Industrial Gyroscope Applications
- 2.3 Motor Speed Control Strategy Selection
- 2.4 PID Control and Tuning Methods
- 2.5 BLDC Motor Technology and ESC Design
- 2.6 Flywheel Design, Material Selection, and Dynamic Balancing
- 2.7 Microcontroller and Sensor Selection
- 2.8 Real-Time Monitoring and Communication Protocols
- 2.9 Database and Persistent Data Storage
- 2.10 Summary and Justification of Chosen Methods

CHAPTER 3: METHODOLOGY

- 3.0 Methodology
- 3.1 Component Design
 - 3.1.1 Flywheel Design
 - 3.1.2 Shaft Design
 - 3.1.3 Bearing Selection
 - 3.1.4 Motor and ESC Selection
 - 3.1.5 Coupling Design
 - 3.1.6 Frame Design

- 3.2 Component Analysis
 - 3.2.1 Design Calculations
 - 3.2.2 Bearing Life Calculation
 - 3.2.3 Thermal Analysis
 - 3.2.4 Structural Analysis of Main Shaft
 - 3.2.5 Control System Design
 - 3.2.6 Software and Monitoring System Architecture
 - 3.2.7 MATLAB Simulink Simulation
 - 3.2.8 Circuit Design and Simulation

CHAPTER 4: MANUFACTURE AND ASSEMBLY

- 4.1 Manufacturing Plan and Bill of Materials
- 4.2 Flywheel Machining
- 4.3 Dynamic Balancing Procedure
- 4.4 Frame Fabrication and Modification
- 4.5 Mechanical Assembly
- 4.6 Electronic Assembly
- 4.7 Firmware Programming and Initial Calibration
- 4.8 Challenges Encountered During Manufacturing

CHAPTER 5: TESTING AND RESULTS

- 5.1 Test Strategy and Safety Plan
- 5.2 Test 1: Motor Control and ESC Characterisation
- 5.3 Test 2: IMU Sensor Accuracy and Noise
- 5.4 Test 3: PID Controller Performance
- 5.5 Test 4: Dashboard Latency and Safety Response
- 5.6 Test 5: 24-Hour Endurance Run
- 5.7 Comparison with Simulation Predictions
- 5.8 Results Summary
- 5.9 Discussion of Results

CHAPTER 6: CONCLUSIONS

- 6.1 Conclusions
- 6.2 Recommendations
- 6.3 Future Work

REFERENCES

APPENDICES

LIST OF FIGURES

Figure	Description
Figure 1.0	Full gyroscope assembly -- diametric view (SOLIDWORKS, 2026)
Figure 1.1	Full gyroscope assembly -- side view (SOLIDWORKS, 2026)
Figure 2.0	Conceptual design -- diametric view (SOLIDWORKS, 2026)
Figure 2.1	Conceptual design -- side view (SOLIDWORKS, 2026)
Figure 2.2	Flywheel engineering drawing with all dimensions (SOLIDWORKS, 2026)
Figure 2.3	Internal component layout -- side section view (SOLIDWORKS, 2026)
Figure 2.4	Shaft and bearing sub-assembly exploded view
Figure 2.5	Spider coupling -- exploded view
Figure 3.0	Free body diagram: gyroscopic torque and precession direction
Figure 3.1	Mass distribution diagram for moment of inertia calculation
Figure 3.2	Bearing load diagram showing radial and axial load components
Figure 3.3	Thermal model block diagram for motor winding temperature
Figure 3.4	Shaft bending moment diagram under imbalance load
Figure 4.0	MATLAB Simulink -- complete system block diagram
Figure 4.1	Simulink PID controller subsystem block diagram
Figure 4.2	Uncontrolled vs. controlled simulation response comparison
Figure 4.3	PID step response -- rise time, settling time, overshoot annotated
Figure 4.4	Simulink motor electrical model subsystem
Figure 5.0	Circuit schematic -- full system (Proteus, 2026)
Figure 5.1	ESC gate drive timing diagram from Proteus simulation
Figure 5.2	Monitoring dashboard -- live RPM chart and stat cards
Figure 5.3	Dashboard -- system events and alert history panel
Figure 6.0	Dynamic balancing rig with flywheel mounted
Figure 6.1	Balance correction hole positions -- before and after annotated
Figure 7.0	CNC turning operation -- flywheel on lathe
Figure 7.1	Completed perf-board electronic assembly -- top view
Figure 7.2	Completed perf-board -- underside showing power trace reinforcement
Figure 8.0	Assembled gyroscope prototype -- full view
Figure 8.1	Motor and coupling detail -- spider coupling installed
Figure 9.0	RPM trend over extended stability run (from dashboard export)
Figure 9.1	Tilt X and Tilt Y over extended stability run
Figure 9.2	Temperature trend over extended stability run

Figure 9.3	Vibration level over extended stability run
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LIST OF TABLES

Table	Description
Table 1	Comparison of motor speed control strategies
Table 2	Comparison of PID tuning methods
Table 3	Comparison of motor types for this application
Table 4	Comparison of microcontroller options
Table 5	Comparison of IMU sensor options
Table 6	Flywheel material property comparison
Table 7	Summary of all design calculations
Table 8	Motor electrical parameters
Table 9	Bearing selection parameters and calculated L10 life
Table 10	Thermal analysis inputs and results
Table 11	Shaft structural analysis results
Table 12	Bill of materials with costs
Table 13	Manufacturing process plan
Table 14	Test plan -- all five tests with pass criteria
Table 15	Test 1 results: motor control and ESC characterisation
Table 16	Test 2 results: IMU accuracy
Table 17	Test 3 results: PID controller performance
Table 18	Test 4 results: dashboard latency and emergency stop
Table 19	Test 5 results: extended stability run (2 hours)
Table 20	Comparison of simulation predictions vs. experimental results
Table 21	Full component specifications
Table 22	Risk assessment
Table 23	Communication protocol comparison

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AC	Alternating Current
BLDC	Brushless Direct Current
BOM	Bill of Materials
CAD	Computer-Aided Design

CNC	Computer Numerical Control
DC	Direct Current
DOF	Degree of Freedom
DRO	Digital Readout
EMF	Electromotive Force
ESC	Electronic Speed Controller
FFT	Fast Fourier Transform
FBD	Free Body Diagram
I2C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
JSON	JavaScript Object Notation
KV	Motor velocity constant (RPM per volt)
LDO	Low-Dropout Regulator
MATLAB	Matrix Laboratory
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MPC	Model Predictive Control
ORM	Object-Relational Mapping
PID	Proportional-Integral-Derivative
PWM	Pulse Width Modulation
REST	Representational State Transfer
RPM	Revolutions Per Minute
SHS	Square Hollow Section
SQL	Structured Query Language
SSE	Server-Sent Events
TCP	Transmission Control Protocol
TIR	Total Indicator Reading
TVS	Transient Voltage Suppressor
USB	Universal Serial Bus
VFD	Variable Frequency Drive
WS	WebSocket
ZN	Ziegler-Nichols

CHAPTER 1: INTRODUCTION

1.1 Background

1.1.1 The Gyroscope -- Historical Development

The gyroscope is one of the most elegant applications of classical mechanics. Its operating principle relies on the conservation of angular momentum, a law that has been understood mathematically since Euler formulated the equations of rigid body motion in 1765. However, the practical rotating-mass gyroscope as an engineering instrument was first described by the French physicist Leon Foucault in 1852, who used a spinning mass to experimentally demonstrate the rotation of the Earth. Foucault coined the term 'gyroscope' from the Greek words for 'circle' and 'to observe' (Hibbeler, 2015, p. 405).

The first commercial applications of gyroscopes emerged in the late 19th and early 20th century. The gyrocompass -- a spinning gyroscope that aligns its spin axis with the Earth's rotation axis -- was developed by Elmer Sperry in 1908 and quickly became the standard navigation instrument for large ships, where magnetic compasses are unreliable due to the ship's own iron structure. The gyrostabiliser for ships followed shortly after, using a large motor-driven flywheel to counteract wave-induced rolling (Meriam, Kraige and Bolton, 2016, p. 476).

Aircraft artificial horizons, which use a gyroscope to maintain a stable reference plane regardless of aircraft attitude, became standard instruments in the 1920s. During World War II, gyroscopes were used in torpedo guidance systems, bombsights, and autopilots, greatly expanding the technology's industrial manufacturing base. By the 1950s, inertial navigation systems using precisely machined gyroscopes were the primary navigation method for intercontinental ballistic missiles and, later, commercial aircraft (Alciatore and Hstand, 2012, p. 445).

The mid-20th century saw the development of electronic alternatives to spinning-mass gyroscopes. Ring laser gyroscopes (RLGs), introduced commercially in the 1970s, use the Sagnac effect -- a phase shift in counter-rotating laser beams caused by rotation -- to measure angular velocity with no moving parts. Fibre-optic gyroscopes (FOGs), developed in the 1980s, use the same physical principle with optical fibre coils. Both offer higher reliability and faster warm-up times than spinning-mass gyroscopes in precision navigation applications. Micro-electromechanical system (MEMS) gyroscopes, the type found in modern smartphones and consumer electronics, use vibrating silicon structures and became commercially available in the early 1990s (Meriam, Kraige and

Bolton, 2016).

Despite the proliferation of electronic alternatives, motorized spinning-mass gyroscopes remain important in applications that benefit from their large stored angular momentum -- particularly flywheel energy storage systems, ship roll stabilisers, satellite attitude control flywheels, and engineering education. The physical intuition of a spinning flywheel resisting tilting forces is also far more accessible to students than the quantum-optical principles of a ring laser gyroscope, making the motorized gyroscope a valuable teaching instrument for dynamics and control systems courses.

1.1.2 Physical Principle of Gyroscopic Stability

The stability of a gyroscope arises from the vector nature of angular momentum. Angular momentum $L = I * \omega$ is a vector directed along the spin axis. By Newton's second law for rotation, the only way to change the angular momentum vector is to apply a torque. An applied torque does not spin up or spin down the gyroscope -- it rotates the spin axis itself, a phenomenon called precession. The larger the angular momentum, the smaller the angle through which the spin axis moves in response to a given torque. A fast-spinning, high-inertia flywheel therefore resists disturbances with very little angular displacement -- it appears 'rigid' in space, which is the property called gyroscopic rigidity (Meriam, Kraige and Bolton, 2016).

For a gyroscope to remain stable under gravity, the gyroscopic reaction torque must exceed the gravitational overturning torque at the maximum expected tilt angle. As speed decreases below a critical minimum, the gyroscopic torque falls, the precession rate increases, and the spin axis begins to nutate and ultimately fall. The target design speed of 3000 RPM in this project corresponds to a stability factor of 21 -- the system is theoretically stable down to approximately 655 RPM, providing a very large margin against speed droop during the endurance run.

1.1.3 The CBU Existing Prototype

A previous student team at CBU designed and built a motorized gyroscope as a final-year engineering project. Their design documentation, which was made available to this team, shows a mild steel flywheel driven by a BLDC motor mounted in a welded steel frame. The flywheel dimensions and mass are comparable to the design used in this project. However, the system had the following documented shortcomings:

1. No dynamic balancing:

The flywheel was machined on a conventional lathe and fitted without any balancing procedure. The previous team's notes record 'severe vibration at high speed' and note that bearings required replacement after less than 10 hours of total operation.

2. Manual speed control only:

A potentiometer connected to the ESC signal wire was adjusted by hand to control motor speed. There was no feedback control. The team's notes state that maintaining a constant speed required 'continuous attention'.

3. Breadboard electronics:

All wiring was on solderless breadboards. The team's log records multiple sessions that had to be terminated early due to sensor wires vibrating loose from breadboard sockets.

4. No monitoring system:

There was no mechanism to monitor RPM, tilt, temperature, or vibration in real time. The only observable indication of system status was the sound of the flywheel and the visual observation of tilt.

5. Operational duration limited:

The longest continuous run recorded in the previous team's log was 3 hours and 41 minutes, after which a breadboard connection failure caused the motor to stop.

[INSERT FIGURE HERE]

Figure 1.0: Full gyroscope assembly -- diametric view (SOLIDWORKS, 2026)

[INSERT FIGURE HERE]

Figure 1.1: Full gyroscope assembly -- side view (SOLIDWORKS, 2026)

1.2 Problem Statement

The existing CBU gyroscope cannot sustain stable, automated operation. The following specific engineering problems have been identified from analysis of the previous team's design documentation and from physical inspection of the prototype:

1. Insufficient operational runtime:

The system cannot maintain stable automated operation. Without automated speed regulation, the motor slows progressively as temperature rises and bearing friction changes. The maximum recorded run time was under 4 hours without manual intervention.

2. Flywheel dynamic imbalance:

The flywheel was not dynamically balanced. Residual imbalance generates centrifugal forces proportional to the square of the rotational speed. At 3000 RPM, even 1 g of imbalance at 100 mm radius generates 44 N of force. This accelerates bearing wear dramatically, reducing operational life from thousands of hours to tens of hours.

3. No real-time monitoring or alerting:

Without sensor data, critical conditions such as overheating, excessive vibration, or speed droop below the stability limit cannot be detected until a failure has already occurred. There is no automatic emergency stop.

4. Manual speed control only:

Potentiometer-based manual speed control cannot maintain a constant setpoint for 24 unattended hours, nor can it respond fast enough to sudden disturbances that would cause the flywheel to fall below the stability limit.

5. Unreliable solderless breadboard electronics:

Breadboard connections rely on friction and are not rated for vibration environments. The previous team experienced repeated connection failures during high-speed operation that could not be reliably reproduced or debugged.

1.3 Aim

To improve, build, test, and validate a cost-effective motorized gyroscope that operates continuously without human intervention, with full real-time monitoring of all critical operating parameters and an automatic safety response system.

1.4 Motivation

1.4.1 Academic Motivation

The project spans the full breadth of both the Mechatronics and Electromechanical Engineering programmes. It simultaneously requires dynamics and mechanics (gyroscope physics, flywheel design, dynamic balancing), control systems (PID design and simulation), electronics (microcontroller programming, sensor integration, ESC design), and software engineering (server, database, dashboard, WebSocket communication). Very few single projects integrate all of these disciplines into one coherent, working, testable product. This made it an exceptionally rich platform for demonstrating the full range of our engineering education.

1.4.2 Professional and Industrial Motivation

Making a machine run reliably for extended periods without human intervention is precisely the kind of challenge faced by professional engineers in industrial maintenance, continuous process plants, and remote monitoring applications. It demands careful design, thorough failure mode analysis, disciplined testing, and the engineering judgment to know when a system is ready for deployment and when it is not. The skills developed through this project -- reliability engineering, condition monitoring, control system design, and embedded systems programming -- are directly relevant to industrial employment.

1.4.3 National Relevance

Gyroscope principles are applied in flywheel energy storage systems, which are of particular interest in Zambia and the wider region as a complement to solar power generation: stored kinetic energy in a flywheel can bridge the gap between periods of intermittent solar generation without the degradation and chemical hazards associated with battery storage. Demonstrating that precision flywheel engineering is achievable within CBU's workshop using locally sourced materials is a direct contribution to the feasibility of such applications in Zambia.

1.5 Objectives

The following specific technical objectives were defined:

1. Design, machine, and dynamically balance a mild steel flywheel capable of stable continuous automated operation above 1500 RPM.
2. Select a motor speed control strategy through a systematic comparative review of all major alternatives.
3. Design, tune, and validate a PID controller for motor speed regulation in MATLAB Simulink before hardware deployment.
4. Build a real-time monitoring system tracking RPM, tilt (X and Y), temperature, vibration, and PWM with configurable safety thresholds and automatic emergency

stop.

5. Replace all breadboard connections with soldered perf-board assembly.
6. Simulate the complete circuit in Proteus before hardware build.
7. Validate the system with an extended stability test with full monitoring active.

1.6 Scope and Limitations

This project covers the complete redesign, fabrication, firmware, software, and testing of a single-axis motorized gyroscope. The frame was inherited from the previous team, constraining the design to single-axis operation. Multi-axis active stabilisation is outside the scope.

All components were sourced within the Copperbelt region, which restricted sensor and motor options. The Proteus simulation environment does not include ESP32 or commercial ESC library models; substitute models were used.

The endurance test was conducted indoors under controlled ambient conditions. The system has not been tested under outdoor environmental conditions (temperature cycling, humidity, dust). Generalisation of results to outdoor deployments would require additional testing.

The RPM measurement method used (vibration FFT analysis) has a resolution limited by the FFT window size. For high-precision applications, a dedicated optical encoder would be required.

1.7 Report Structure

Chapter 2 presents the literature review covering all major design decisions: gyroscope theory, motor speed control strategies, PID tuning methods, BLDC motor and ESC technology, flywheel design and balancing, microcontroller and sensor selection, real-time communication protocols, and database technology. Each section ends with a justified selection.

Chapter 3 covers the full methodology: mechanical design, engineering calculations, bearing life calculation, thermal analysis, structural analysis, control system design, software architecture, Simulink simulation, and circuit design. Chapter 4 covers manufacturing and assembly in detail. Chapter 5 presents all five test procedures and results. Chapter 6 states conclusions, recommendations, and future work directions.

CHAPTER 2: LITERATURE REVIEW

2.1 Gyroscope Theory and Angular Momentum

2.1.1 Conservation of Angular Momentum

Angular momentum is the rotational analogue of linear momentum. For a rigid body rotating about a fixed axis, angular momentum L is defined as (Hibbeler, 2015, p. 412):

$$L = I * \omega_s \quad [kg.m^2/s]$$

where I is the moment of inertia about the spin axis and ω_s is the angular velocity of spin. By Newton's second law for rotation, the net external torque equals the rate of change of angular momentum:

$$\tau = dL/dt$$

If no net external torque acts on the system, angular momentum is conserved -- its magnitude and direction remain constant. This means the spin axis maintains its orientation in space, regardless of how the support structure moves around it. This property is called gyroscopic rigidity and is the fundamental principle exploited by all gyroscopic instruments (Meriam, Kraige and Bolton, 2016, p. 486).

2.1.2 Gyroscopic Precession

When an external torque T_{external} is applied to a spinning gyroscope, the angular momentum vector L rotates at a rate ω_p called the precession rate. The relationship between the applied torque, angular momentum, and precession rate is (Meriam, Kraige and Bolton, 2016, p. 488):

$$T_{\text{external}} = L \times \omega_p = I * \omega_s * \omega_p$$

Rearranging to give the precession rate produced by a given torque:

$$\omega_p = T_{\text{external}} / (I * \omega_s) \quad [rad/s]$$

This is the key stability relationship: a faster spin speed (larger $I * \omega_s$) produces a smaller precession rate in response to the same external torque. The gyroscope appears more 'rigid' in space as spin speed increases. For stability against gravity, the gyroscopic torque produced at the precession rate must exceed the gravitational overturning torque:

$$I * \omega_s * \omega_p \geq m * g * h * \sin(\theta) \quad (\text{Ogata, 2010, p. 74})$$

This inequality defines the minimum spin speed for stability at any given tilt angle θ . It is the central design criterion for the flywheel.

2.1.3 Nutation

In addition to precession, a disturbed gyroscope exhibits nutation -- a rapid, small-amplitude oscillation of the spin axis superimposed on the slower precession. Nutation frequency is proportional to spin speed and inversely proportional to the transverse moment of inertia (Hibbeler, 2015, p. 418). At 3000 RPM, the nutation frequency of the gyroscope is very high and damps out rapidly due to air resistance and bearing friction, making nutation practically invisible during operation. For the purposes of this project's control system design, nutation was not modelled separately.

2.1.4 Moment of Inertia for a Disk

For a uniform solid disk rotating about its geometric central axis (Hibbeler, 2015, p. 388):

$$I = (1/2) * m * r^2 \quad [kg.m^2]$$

For a disk with mass concentrated towards the rim, the effective moment of inertia is higher for the same total mass and radius. Adding mass at the periphery (for example, with rim bolts) is therefore an efficient strategy for increasing angular momentum without increasing the flywheel diameter or requiring a higher motor speed. This principle was used in the flywheel design for this project.

[INSERT FIGURE HERE]

Figure 3.0: Free body diagram showing gyroscopic torque (T_g), gravitational overturning torque (T_{grav}), and precession direction

[INSERT FIGURE HERE]

Figure 3.1: Mass distribution diagram showing flywheel disk, hub, and rim bolt positions with radii labelled

2.2 Review of Industrial Gyroscope Applications

Understanding how gyroscopes are applied industrially provides context for the design choices made in this project and identifies relevant engineering precedents.

2.2.1 Ship Roll Stabilisers

Active gyro stabiliser systems for ships use large motor-driven flywheels mounted on gimbals. When the ship rolls, the gyroscope precesses, and actuators on the gimbal axes apply a counter-torque to the ship's hull, reducing the roll amplitude. Modern ship gyro stabilisers use flywheels spinning at 1500-11000 RPM with masses from 2 tonnes (small yachts) to over 200 tonnes (large cruise ships). The active control system monitors the ship's roll angle in real time and adjusts the gimbal angle to maximise the stabilising torque. This is a scaled-up version of exactly the monitoring and control system built in this project (Meriam, Kraige and Bolton, 2016, p. 490).

2.2.2 Flywheel Energy Storage

Flywheel energy storage systems (FESS) store kinetic energy in a spinning mass and release it on demand. Modern FESS units spin carbon fibre composite rotors at up to 60000 RPM in a vacuum enclosure supported on active magnetic bearings, storing 1-100 kWh per unit. They are used for grid frequency regulation and as uninterruptible power supplies for data centres. The engineering principles -- high-speed rotor dynamics, dynamic balancing, bearing selection, and speed control -- are identical to those applied in this project, though at greatly different scales.

2.2.3 Satellite Reaction Wheels

Satellite attitude control uses three orthogonal reaction wheels -- motor-driven flywheels mounted on frictionless magnetic bearings. By accelerating or decelerating each wheel, the satellite imparts an equal and opposite angular impulse to the satellite body, rotating it

to point instruments or antennas in the desired direction. The requirements for reaction wheels -- high speed, zero maintenance, precise speed control, and real-time telemetry -- are directly analogous to the requirements for this project's gyroscope system, though the operating environment is vastly different.

2.3 Motor Speed Control Strategy Selection

Maintaining a 3000 RPM setpoint for 24 unattended hours requires automatic feedback control. As temperature and bearing friction change slowly over hours of operation, and as external disturbances occasionally perturb the flywheel, the control system must detect speed deviations and correct them automatically. Four principal control strategies were evaluated:

2.3.1 ON/OFF (*Bang-Bang*) Control

ON/OFF control switches the motor fully on when speed is below the setpoint and fully off when it is above (Ogata, 2010, p. 64). It is the simplest possible feedback control law, requiring only a comparator with no mathematical model. The fundamental drawback is that the controller can never reach a true steady state -- it always overshoots in one direction, then switches and overshoots in the other, producing continuous oscillation around the setpoint. For a spinning flywheel, where speed oscillation directly impacts gyroscopic rigidity and mechanical stress, this behaviour is fundamentally unsuitable. ON/OFF control was rejected without detailed analysis.

2.3.2 PID Control

A PID controller computes its output from three terms: proportional (P), integral (I), and derivative (D) of the error $e(t) = \text{setpoint} - \text{measured speed}$ (Astrom and Murray, 2010, p. 74):

$$u(t) = K_p * e(t) + K_i * (\text{integral of } e \text{ } dt) + K_d * (de/dt)$$

The proportional term provides immediate response to current error. The integral term eliminates steady-state error by accumulating total error over time -- without it, the controller would settle at an offset from the setpoint. The derivative term acts as a predictive damper, reducing overshoot by reacting to the rate of change of error before it becomes large.

PID has been the dominant industrial control algorithm since the 1940s. It requires no prior mathematical model of the controlled system. The gain values can be tuned systematically. For the approximately linear relationship between PWM duty cycle and motor speed, a well-tuned PID achieves fast response with minimal overshoot and zero steady-state error (Astrom and Murray, 2010, p. 83).

2.3.3 Fuzzy Logic Control

Fuzzy logic control uses linguistic rules of the form 'IF error is Large THEN increase output by Much' rather than a mathematical control law. It can handle inherently nonlinear systems and does not require a system model. The primary disadvantage for this project is the complexity of designing and validating the rule base: the system designer must enumerate and weight all relevant cases, which requires extensive manual iteration. Since the BLDC-flywheel system is approximately linear at the operating point, fuzzy logic offers no meaningful advantage over PID and was not chosen (Alciatore and Hristand, 2012, p. 514).

2.3.4 Model Predictive Control

MPC solves an optimisation problem at each control step, minimising a cost function over a future prediction horizon while respecting input and output constraints. It is the most powerful of the four strategies reviewed and handles constraints and multi-variable interactions elegantly (Astrom and Murray, 2010, p. 312). However, MPC requires a detailed system model, careful tuning of the prediction horizon and weighting matrices, and substantially more computational resources. Implementing MPC on an ESP32 at a 50 Hz update rate, while also running the IMU, FFT, and communications tasks, would be very challenging. MPC was rejected on the grounds of implementation complexity relative to benefit.

Table 1: Comparison of Motor Speed Control Strategies

Strategy	SS Error	Overshoot	Model?	Complexity	Suitable?
ON/OFF	High (oscillates)	Very high	No	Very low	No
PID	Zero (with I)	<10%	No	Low	Yes -- CHOSEN
Fuzzy Logic	Low	Low	No	Medium	No
MPC	Zero	Very low	Yes	High	No

PID was selected for its zero steady-state error, well-established tuning procedures, low embedded implementation complexity, and extensive industrial track record for motor speed regulation.

2.4 PID Control and Tuning Methods

Selecting PID as the control strategy raises the question of how to choose the three gain values K_p , K_i , and K_d . These parameters control the speed, damping, and accuracy of the closed-loop response.

2.4.1 Ziegler-Nichols Open-Loop Method

The classical Ziegler-Nichols (ZN) method uses the system open-loop step response to identify three parameters: process gain K , dead time L , and dominant time constant T . Gains are then calculated from (Ogata, 2010, p. 231):

$$K_p = 1.2 \cdot T / (K \cdot L), \quad K_i = K_p / (2 \cdot L), \quad K_d = 0.5 \cdot K_p \cdot L$$

The ZN method typically produces gains that are aggressive -- leading to 20-30 percent overshoot and oscillatory response that usually requires subsequent manual detuning. For a high-speed flywheel, large transient speed excursions are mechanically undesirable. ZN was used as a starting point for simulation comparison but not as the primary tuning method.

2.4.2 Ziegler-Nichols Closed-Loop Method

The ZN closed-loop (ultimate gain) method requires the physical system to be driven to the boundary of stability -- marginal oscillation. This procedure carries clear mechanical risk for a high-speed flywheel. The method was considered and rejected on safety grounds.

2.4.3 Cohen-Coon Method

The Cohen-Coon method (cited in Alciatore and Hinand, 2012, p. 442) uses the same step response data as ZN open-loop but different gain formulas developed to provide better load disturbance rejection. It produces moderately aggressive gains and, like ZN, typically requires subsequent fine-tuning. It was noted as an alternative to ZN but not selected as the primary method for this project.

2.4.4 Manual (Empirical) Tuning

Manual tuning follows a structured sequence: set $K_i = K_d = 0$ and increase K_p until the closed-loop response is acceptably fast; then add K_i to eliminate steady-state error; then add K_d to reduce overshoot. The method is practical and widely used but produces gains that cannot be justified analytically without simulation. Manual tuning was used for final small adjustments after hardware deployment.

2.4.5 MATLAB Simulink Auto-Tuner

MATLAB Simulink PID Tuner (MathWorks, 2026) uses the linearised plant model from

the Simulink block diagram to automatically calculate PID gains that meet user-specified targets for closed-loop bandwidth (related to response speed) and phase margin (related to damping and overshoot). All tuning is done in simulation -- there is zero hardware risk. The resulting gains can be tested against multiple disturbance and setpoint change scenarios in simulation before being transferred to the microcontroller. This method is strongly preferred when a simulation model is available, as it integrates directly with the Simulink design workflow.

Table 2: Comparison of PID Tuning Methods

Method	HW Risk	Typical Overshoot	Justified?	Selected?
ZN Open-Loop	Low	20-30%	Yes	Comparison only
ZN Closed-Loop	High	20-30%	Yes	No -- unsafe
Cohen-Coon	Low	Moderate	Yes	No
Manual Tuning	Low	Varies	No	Final adj. only
MATLAB Auto-Tuner	None	<10%	Yes	Yes -- PRIMARY

2.5 BLDC Motor Technology and ESC Design

2.5.1 Brushed vs. Brushless DC Motors

Brushed DC motors use physical carbon brushes in contact with a rotating commutator to deliver current to the rotor windings. Brush-commutator friction generates heat, carbon dust, and electrical arcing. Brush life is typically 1000-3000 hours under moderate load, with wear rate increasing sharply at higher speeds (Alciatore and Hiestand, 2012, p. 232). At 3000 RPM during sustained operation, brush replacement mid-run is a credible failure mode.

Brushless DC motors eliminate physical commutation by moving the windings to the stator and using permanent magnets on the rotor. Current switching between stator windings is handled electronically by the ESC. With no brush contact, there is no mechanical wear in the motor itself -- the only wear surfaces are the motor bearings, which are rated for tens of thousands of hours at moderate load. BLDC motors also achieve higher efficiency (typically 85-92 percent) than brushed DC (70-75 percent),

resulting in less heat generation at the same mechanical output (Alciatore and Hristand, 2012, p. 241).

2.5.2 KV Rating and Torque Considerations

The KV rating of a BLDC motor is the no-load speed in RPM per volt of supply voltage. For a 950 KV x2212 motor, the no-load speed at 3.64 V is approximately 3000 RPM. Higher KV motors have lower torque constants and are therefore better suited to low-load, high-speed applications. A lower-KV motor (for example 500 KV) would produce more torque at the same current but would require a higher supply voltage to reach 3000 RPM and would be physically larger and heavier for the same power rating. The 950 KV rating was selected because: at 12 V supply and 36% ESC duty cycle, the effective winding voltage of 4.36 V delivers 3000 RPM no-load speed; the higher torque constant (0.01005 N.m/A vs. 0.00434 for a 2200 KV motor) provides better low-speed starting margin for the flywheel; and the x2212 stator size gives an efficient operating point at this power level.

2.5.3 ESC Design

Commercial hobby-grade BLDC ESCs accept a 50 Hz PWM signal with pulse width between 1000 and 2000 microseconds. A 1000 us pulse commands minimum throttle; 2000 us commands maximum. The ESC internally runs a six-step trapezoidal commutation sequence, sensing rotor position from the back-EMF of the undriven winding. The hardware design for the custom ESC built in this project follows the standard topology: three half-bridge circuits using power MOSFETs, gate driver ICs for high-side and low-side switching, and current sensing for protection.

Table 3: Comparison of Motor Types for this Application

Motor Type	Brush Wear?	Efficiency	Speed Range	Long-run Suitability
Brushed DC	Yes	70-75%	Low-medium	Not suitable
BLDC	No	85-92%	Wide	Excellent for sustained operation -- CHOSEN

Stepper motor	No	50-70%	Low	Not suitable -- too slow
AC induction	No	80-90%	Fixed	Possible but needs VFD

2.6 Flywheel Design, Material Selection, and Dynamic Balancing

2.6.1 Material Selection

The flywheel material selection was based on four criteria: density (higher density = more angular momentum in a given volume), machinability (must be machinable in CBU workshop), ductility (brittle fracture at speed is a safety hazard), and local availability and cost.

Table 6: Flywheel Material Property Comparison

Property	Mild Steel	Aluminium 6061	Cast Iron
Density (kg/m ³)	7850	2700	7200
Tensile strength	400 MPa	310 MPa	200 MPa
Elongation at break	25%	17%	<1% (brittle)
Machinability	Good (CNC)	Excellent	Fair
Local cost	Low	Medium	Low
Density advantage	Highest	Low -- 3x less	Similar to steel
Fracture mode	Ductile	Ductile	Brittle (unsafe)
Selected?	YES	No	No -- brittle

Mild steel was selected. Its high density concentrates angular momentum in a compact volume, minimising the flywheel diameter. Aluminium would require a 70 percent larger diameter flywheel for the same inertia. Cast iron was rejected for its brittle fracture behaviour.

2.6.2 Dynamic Balancing Theory

A rotating body has dynamic imbalance when its principal axis of inertia does not coincide with the rotation axis. This creates a rotating couple that generates oscillating loads on the bearings at the frequency of rotation. The magnitude of the centrifugal force from an imbalance mass m at eccentricity e , rotating at angular velocity ω , is (Meriam, Kraige and Bolton, 2016, p. 420):

$$F_c = m * e * \omega^2$$

At 3000 RPM ($\omega = 314.16$ rad/s), 1 g of imbalance at 100 mm radius generates $F_c = 0.001 * 0.1 * 314.16^2 = 70.1$ N -- a force that cycles 133 times per second (at 3000 RPM) and would fatigue ball bearings within hours. ISO 1940 Grade G6.3 allows a maximum specific unbalance of 6.3 g.mm/kg. For a 0.5 kg flywheel, the maximum allowed residual unbalance is 5.4 g.mm, producing a centrifugal force of only 0.004 N at operating speed.

2.6.3 Two-Plane Balancing

Static balancing eliminates imbalance in a single transverse plane. Dynamic balancing (two-plane balancing) eliminates imbalance in two longitudinal planes, correcting both the net centrifugal force and the imbalance couple. This requires a balancing rig with vibration sensors in two measurement planes and a phase reference sensor. The influence coefficient method calculates the correction mass and angular position for each plane from two sets of vibration measurements -- one with a known trial mass added, one without (Meriam, Kraige and Bolton, 2016, p. 422). This is the method used in Section 4.3.

2.7 Microcontroller and Sensor Selection

2.7.1 Microcontroller

The microcontroller requirements: minimum 32-bit architecture for floating point PID calculations, clock speed ≥ 100 MHz for real-time task scheduling at 50 Hz, ≥ 100 KB RAM for IMU data buffers and FFT workspace, and built-in Wi-Fi for wireless monitoring. The evaluation covered five candidate boards:

Table 4: Comparison of Microcontroller Options

Board	Architecture	RAM	Wi-Fi?	Cost	Suitable?
Arduino Uno	8-bit/16 MHz	2 KB	No	Low	No -- too slow

Arduino Mega	8-bit/16 MHz	8 KB	No	Low	No -- no Wi-Fi
STM32F4	32-bit/168 MHz	192 KB	No	Med	Possible -- no Wi-Fi
Raspberry Pi Zero 2	32-bit/1 GHz	512 MB	Yes	Med	Overkill; OS overhead
ESP32 Xtensa LX6	32-bit/240 MHz	520 KB	Yes	Low	YES -- chosen

The ESP32 was selected for its combination of dual-core 240 MHz processing power, 520 KB SRAM, and integrated Wi-Fi/Bluetooth on a single chip. Its dual-core architecture allows communications tasks to run on Core 0 without interrupting the real-time control loop on Core 1 (Maier, Sharp and Vagapov, 2017).

2.7.2 IMU Sensor

The IMU must provide 3-axis tilt measurements and accelerometer data suitable for vibration FFT analysis. The required accuracy is +/-1 degree for tilt and a sampling rate of at least 100 Hz for vibration. Three options were evaluated:

Table 5: Comparison of IMU Sensor Options

Sensor	Axes	Interface	Resolution	Cost	Selected?
ADXL345	3-axis accel	I2C/SPI	13-bit	Low	No -- no gyro
MPU6050	6-axis a+g	I2C	16-bit	Low	YES
LSM9DS1	9-axis a+g+mag	I2C/SPI	16-bit	High	No -- not local

The MPU6050 provides both a 3-axis accelerometer and a 3-axis gyroscope in one package. The combination enables complementary or Kalman filtering for accurate, vibration-resistant tilt estimates that neither sensor alone could provide. The 16-bit ADC gives 0.0305 deg/s resolution in the +/-250 deg/s range, more than adequate for tilt measurement. It is also inexpensive and widely available (InvenSense, 2013).

2.8 Real-Time Monitoring and Communication Protocols

2.8.1 HTTP Polling

In an HTTP polling architecture, the browser client sends a GET request to the server at regular intervals (typically every 1-5 seconds) to retrieve the latest sensor data. This is easy to implement but has fundamental latency limitations: data is at most as fresh as the polling interval. A 2-second interval means a threshold violation may not be visible for up to 2 seconds -- unacceptable for a safety monitoring system. HTTP polling also generates a constant stream of network requests regardless of whether the data has changed.

2.8.2 Server-Sent Events

Server-Sent Events (SSE) allow the server to push data to the browser over a persistent HTTP connection whenever new data is available, eliminating polling delay. However, SSE is unidirectional: only the server can push to the client. Motor control commands cannot be sent back from the dashboard to the server over the same SSE connection, requiring a separate HTTP request for each control action. For a dashboard that both displays data and sends motor commands, SSE alone is not sufficient.

2.8.3 WebSocket

WebSocket (RFC 6455) establishes a persistent, full-duplex TCP connection between the browser and the server after an HTTP upgrade handshake. Both parties can send messages at any time without overhead of a new HTTP request. The server pushes new sensor readings immediately when they arrive from the ESP32, typically with end-to-end latency under 100 ms on a local network (MDN Web Docs, 2024). The same connection carries motor control commands from the dashboard to the server.

Table 23: Communication Protocol Comparison

Protocol	Latency	Bi-directional?	Complexity	Selected?
HTTP Polling	High (poll interval)	Yes (separate)	Very low	No
SSE	Low (push)	No (server->client only)	Low	No
WebSocket	Very low (<100 ms)	Yes (full-duplex)	Medium	YES

WebSocket was selected as the standard for real-time industrial monitoring dashboards. It

provides full-duplex low-latency communication, and Node.js's ws library makes it straightforward to implement on the server.

2.9 Database and Persistent Data Storage

A persistent database is required to store sensor readings, session records, and alert events for post-run analysis and reporting. Three options were considered: SQLite (file-based relational database), PostgreSQL (client-server relational database), and MongoDB (document store).

SQLite is simple and requires no server process, but has limited concurrent write performance and is not suitable for production deployments with high write rates (such as 50 Hz sensor data). MongoDB is flexible but its schema-less design makes it harder to enforce data integrity for the structured sensor readings used here.

PostgreSQL was selected for its robustness, structured schema support, strong concurrent write performance, and excellent ecosystem of client libraries. Drizzle ORM was used to define the database schema in TypeScript, providing type-safe database access and automatic migration management. All readings are stored at the ingest rate and available for export and analysis after the endurance run.

2.10 Summary and Justification of Chosen Methods

The literature review evaluated all major design decisions systematically. The selected methods for each decision are summarised below.

Design Decision	Method Chosen	Key Justification
Control strategy	PID control	Simple; no model; zero SS error
PID tuning	MATLAB Simulink Auto-Tuner	No HW risk; integrates with simulation
Motor type	BLDC (950 KV x2212)	No brush wear; high efficiency; ideal for sustained operation

Flywheel material	Mild steel	High density; ductile; CNC-machinable
Flywheel mfg.	CNC turning	Best geometric accuracy; minimum initial imbalance
Balancing	Dynamic 2-plane (ISO G6.3)	Eliminates static and dynamic imbalance
Microcontroller	ESP32 dual-core	Wi-Fi built-in; 240 MHz; sufficient RAM
IMU sensor	MPU6050	6-axis; I2C; affordable; locally available
Real-time comms	WebSocket (RFC 6455)	Full-duplex; instant push; <100 ms latency
Server runtime	Node.js + Express 5	Non-blocking I/O; good WebSocket support
Database	PostgreSQL + Drizzle ORM	Robust; type-safe; structured data
Dashboard	React + Recharts	Component-based; live chart support

CHAPTER 3: METHODOLOGY

3.0 Methodology

Design Approach and Engineering Process

The design followed a structured iterative engineering methodology. The process was divided into six sequential phases, each with defined inputs, outputs, and review criteria:

1. Literature review: evaluate all major design decisions and document justified selections (Chapter 2).
2. Systems-level design: define all subsystems, their interfaces, data flows, and performance requirements. Identify failure modes at the system level.
3. Detailed component design: design each subsystem in detail including engineering calculations, material specifications, and dimensional tolerances.
4. Simulation: verify the control system design in MATLAB Simulink and the circuit design in Proteus before any hardware is built or ordered.
5. Manufacture and assembly: fabricate all mechanical components, assemble electronics, and program firmware following the simulation-verified designs.
6. Test and validate: execute the five-stage test programme, comparing results to simulation predictions.

[INSERT FIGURE HERE]

Figure 2.0: Conceptual design -- full assembly diametric view (SOLIDWORKS, 2026)

[INSERT FIGURE HERE]

Figure 2.1: Conceptual design -- side view showing internal layout

System-Level Architecture

The complete system consists of five major subsystems that communicate through defined interfaces:

1. Mechanical subsystem:

Flywheel, shaft, bearings, coupling, and frame. Converts motor torque to spinning angular momentum. Transfers loads to the frame structure.

2. Motor and ESC subsystem:

950 KV x2212 BLDC motor, custom 3-phase ESC, and 12 V power supply. Converts electrical power to mechanical torque in response to PWM commands.

3. Sensing and control subsystem:

ESP32 microcontroller, MPU6050 IMU, Kalman filter, FFT vibration analysis, and PID controller. Measures system state and computes motor commands.

4. Communications subsystem:

USB serial bridge and Wi-Fi TCP socket. Transfers telemetry from the ESP32 to the server and control commands from the server to the ESP32.

5. Server and dashboard subsystem:

Node.js Express server, PostgreSQL database, WebSocket server, and React dashboard. Receives telemetry, evaluates safety thresholds, stores data, and provides the operator interface.

3.1 Component Design

This section presents the design of each mechanical and electrical component of the gyroscope, in the order they were selected: flywheel, shaft, bearings, motor and ESC, coupling, and frame.

3.1.1 Flywheel Design

The flywheel is the most critical mechanical component. Its moment of inertia, combined with its operating speed, determines the angular momentum and hence the gyroscopic stability of the system.

3.1.1.1 Design Requirements

The flywheel design requirements were:

- Moment of inertia $I \geq 0.004 \text{ kg.m}^2$ (from stability analysis)
- Geometrically symmetric to minimise initial dynamic imbalance
- Machinable from locally available mild steel stock on CBU's CNC lathe
- Balance achievable to ISO G6.3 grade using the CBU balancing rig
- Structural integrity with safety factor ≥ 2.0 at 10000 RPM (25% overspeed)
- Mass $\leq 0.8 \text{ kg}$ to remain within the bearing load capacity calculations

3.1.1.2 Geometry

The flywheel was designed as a solid mild steel disk: diameter 215.8 mm, thickness 20 mm. Eight peripheral bolts were added to increase the effective moment of inertia: four M12 bolts at radius 99 mm and four M6 bolts at radius 95 mm. The bolts are indexed at 90-degree spacing (M12) and 45-degree spacing (M6) to maintain geometric symmetry. The hub bore (8 mm) is an interference fit on the shaft; a set screw provides secondary retention. Total design mass including bolts: 0.5 kg.

[INSERT FIGURE HERE]

Figure 2.2: Flywheel engineering drawing -- all dimensions in mm

[INSERT FIGURE HERE]

Figure 3.1: Mass distribution diagram showing bolt positions and radii used in inertia calculation

3.1.2 Shaft Design

The main shaft is EN3B mild steel, 8 mm diameter, 120 mm long. The shaft connects the motor coupling hub at one end to the flywheel hub at the other, and is supported by two RS608 ball bearings located 35 mm from each end.

The shaft diameter was not sized independently; it was set to 8 mm to match the bore of the RS608 bearings already fixed by the inherited frame geometry (Section 3.1.3), since re-boring the existing bearing housings to accept a larger shaft and bearing pair would have required rebuilding the frame, outside the scope of this improvement project. EN3B mild steel was chosen over an alloy steel (e.g. EN8) because it is readily available from CBU's workshop stock and easier to turn to the bearing-fit tolerance on the available lathes; the structural analysis in Section 3.2.4 later confirmed a combined safety factor of 49.6 at the design load, an order of magnitude beyond what the load case required, so the lower-cost, easier-to-machine material was the appropriate choice.

The bearing spacing of 50 mm was determined by the frame geometry inherited from the previous team. A wider spacing would reduce shaft deflection under load, but the existing bearing housings could not be moved without rebuilding the frame. The 50 mm span was verified adequate in the structural analysis (Section 3.2.4), which found a bending stress of only 1.31 MPa against a safety factor of 328.

[INSERT FIGURE HERE]

Figure 2.4: Shaft and bearing sub-assembly -- exploded view showing shaft, inner and outer bearing races, and flywheel hub

3.1.3 Bearing Selection

The bearing bore was fixed at 8 mm by the shaft diameter (Section 3.1.2), which narrowed the search to standard 8 mm-bore deep groove ball bearings available through CBU's regional suppliers. RS608 shielded deep groove ball bearings were selected over an open (unshielded) equivalent and a 2RS double rubber-sealed variant: the shielded type is factory-packed with sufficient grease for its rated life without the extra running friction of contact seals, and it was the variant in local stock without an import lead time the project's timeline could not absorb. Key parameters:

Parameter	Value
Bore diameter	8 mm
Outer diameter	22 mm
Width	7 mm
Dynamic load rating C	3450 N
Static load rating C0	1560 N
Max speed (grease lube)	32000 RPM
Operating temperature	-20 to +120 degC
Lubrication	Grease -- sealed, no maintenance

The dynamic load rating was verified adequate, not merely assumed, in the bearing life calculation of Section 3.2.2: at the applied radial load of 2.45 N and an operating speed of 3000 RPM, the calculated L10 life is 85,900 hours -- confirming bearing failure is not a credible failure mode for this application. This margin reflects standardising on a single stocked, no-maintenance bearing size shared with the shaft-bore constraint above, rather than introducing a second bearing purely to trim an already inexpensive component.

3.1.4 Motor and ESC Selection

3.1.4.1 Motor Selection

The 950 KV x2212 BLDC motor was chosen based on the speed-voltage relationship $n = KV * V$, calculated in Section 3.2.1. At the design supply voltage of 4.36 V, the no-load speed is approximately 3000 RPM.

KV was not the only free variable: for a fixed 3000 RPM target, a lower-KV motor needs a higher supply voltage and a higher-KV motor needs a lower one, while the torque constant k_t (available torque per amp) scales inversely with KV. Table 7a compares the selected 950 KV x2212 motor against two commercially available alternatives at the same target speed, using these relationships.

Table 7a: Motor KV Selection -- Trade-off at Fixed 3000 RPM Target

Motor Option	Voltage for 3000 RPM	Relative k_t	Suitability
1400 KV	~2.14 V eff. (18% duty @ 12V)	Lower (~0.68x)	Rejected -- very low effective voltage; reduced torque constant cuts starting margin
950 KV x2212 (chosen)	4.36 V	Baseline (0.01005 N.m/A)	Selected -- matches the available supply and gives adequate stall torque (0.1005 N.m at 10 A)

600 KV	~5.0 V eff. (50% duty @ 12V)	Higher (~1.68x)	Rejected -- physically larger and heavier; overkill torque for this flywheel load at 3000 RPM
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Full electrical parameters for the selected motor:

Parameter	Value
KV rating	2200 RPM/V
No-load current I_0	1.5 A
Maximum current	10 A
Stator resistance R	0.12 ohm
Number of poles	12 (6 pole pairs)
Mass	45 g
Shaft diameter	3 mm

Table 8: Motor Electrical Parameters

Parameter	Value
Back-EMF constant k_e	0.01005 V.s/rad
Torque constant k_t	0.01005 N.m/A
Stall torque	0.1005 N.m at 10 A
Running torque	~0.030 N.m at 3 A steady-state
Electrical power	43.6 W at full load
Mechanical efficiency	91.6% (steady-state)

3.1.4.2 ESC Design and Selection

The flywheel motor is driven by a commercial, off-the-shelf 30 A BLDC ESC rather than a purpose-built driver stage. A custom three-half-bridge driver (IRFZ44N MOSFETs with IR2101 gate drivers) was modelled in Proteus during the control-system simulation work (Section 3.2.5) because Proteus has no library part for a commercial ESC, but that custom circuit was a simulation-only stand-in -- it was never built into the physical prototype. The physical machine uses the commercial ESC described below throughout.

Key specifications, from the manufacturer datasheet:

Parameter	Value
Continuous / peak current	30 A continuous; 40 A for 10 s
Input voltage range	2-4S LiPo/Li-ion (7.4-14.8 V) or 5-12 cell NiMH/NiCd
BEC output	5 V, 3 A
Max electrical speed (12-pole motor)	35000 RPM
Control signal	Standard 50-60 Hz PWM, 1-2 ms pulse width
Protections	Low-voltage cut-off, over-heat cut-back above 110C, throttle-signal-loss cut-back, abnormal-start cut-off
Mass / size	32 g / 55 x 26 x 13 mm

Table 8a: Commercial ESC Specifications

This ESC was selected over building a custom driver for the physical prototype for four reasons. First, its 30 A continuous / 40 A surge rating is far beyond the motor's own 10 A maximum current draw and ~3 A steady-state running current, so it operates well inside its own thermal and electrical limits under every load case this project produces -- a hand-built driver would need to be engineered and verified to reach the same margin for no operational benefit. Second, its built-in protections (over-temperature cut-back, low-voltage cut-off, throttle-signal-loss cut-back) directly satisfy the safety requirements the project already needed, without adding them in firmware or extra hardware. Third, it accepts the same standard 50-60 Hz, 1-2 ms PWM signal already generated by the ESP32 LEDC peripheral for the PID output stage (Section 3.2.5), so it interfaces directly with no additional signal conditioning. Fourth, as a pre-tested commercial unit it removed a point of failure risk from the build schedule that a hand-built MOSFET/gate-driver stage would have carried, which is why the custom design was kept to a simulation-only role instead.

Because a commercial ESC drives the motor by pulse-width-modulating the raw supply rail rather than physically stepping the supply voltage down, the 4.36 V figure used in the speed calculation of Section 3.2.1 is the effective average voltage delivered to the motor windings at the operating throttle command, not the ESC's raw input voltage. The physical prototype's raw supply rail is a bench AC-DC adapter (110-240 VAC input; selectable 12/15/16/18/19 V at 4.5 A or 20/24 V at 5 A DC output) set to its 12 V tap, chosen because it falls within the ESC's rated 2-4S LiPo / NiMH-NiCd input window and was available on hand in place of a dedicated battery pack. At a 12 V supply rail, the ESP32's PWM throttle command was tuned to a duty cycle of approximately 40% (4.36 V / 12 V) to produce the 3000 RPM no-load operating speed assumed throughout the

design calculations. If a different supply voltage is used, this duty cycle must be re-tuned to keep the no-load speed -- and therefore the stability, inertia, and power calculations of Section 3.2.1 -- valid.

Verification of the throttle interface and current draw against the datasheet's ratings is reported experimentally in Section 5.2 (Test 1: Motor Control and ESC Characterisation).

3.1.5 Coupling Design

A rubber spider (jaw) coupling connects the motor shaft to the flywheel shaft, replacing the rigid, in-line coupling used in the previous prototype. Three alternatives were considered against the requirement to isolate motor commutation vibration from the precision-balanced flywheel shaft while tolerating the small misalignment inevitable in a hand-assembled frame:

- Rigid (sleeve/set-screw) coupling -- the previous design. Rejected: transmits vibration and shock directly into the flywheel shaft and bearings, and has zero misalignment tolerance.
- Bellows coupling -- precise, near-zero backlash. Rejected: designed for torsional stiffness and positioning accuracy, not vibration damping, so it would not address the commutation-vibration problem.
- Elastomeric spider (jaw) coupling -- chosen. The elastomeric insert absorbs shock during motor engagement with the already-spinning flywheel, isolates commutation vibration from reaching the flywheel shaft and bearings, and tolerates up to 1 degree angular and 0.2 mm parallel misalignment.

[INSERT FIGURE HERE]

Figure 2.5: Spider coupling -- exploded view showing motor hub, elastomeric insert, and flywheel hub

3.1.6 Frame Design

The steel frame was inherited from the previous team and modified. Modifications: bearing housings redrilled and reamed to accept RS608 bearings; motor mounting plate

reinforced with 25 x 3 mm flat bar bracing; all surfaces wire-brushed and repainted.

[INSERT FIGURE HERE]

Figure 2.3: Internal component layout -- side section view showing frame, bearings, shaft, motor, and coupling positions

3.2 Component Analysis

This section verifies the components selected in Section 3.1 through engineering calculation, and presents the control system, software, and simulation work that underpins the design.

3.2.1 Design Calculations

3.2.1.1 Moment of Inertia -- Flywheel Disk

For a uniform solid disk (Hibbeler, 2015, p. 388):

$$I_{\text{disk}} = (1/2) * m * r^2$$

$$I_{\text{disk}} = (1/2) * 0.5 * (0.1079)^2$$

$$I_{\text{disk}} = (1/2) * 0.5 * 0.011642$$

$$I_{\text{disk}} = 0.25 * 0.011642 = 0.002910 \text{ kg.m}^2$$

3.2.1.2 Additional Inertia from Rim Bolts

Each bolt is a point mass at its bolt-circle radius:

$$I_{\text{M12}} = 4 * m_{\text{M12}} * r_{\text{M12}}^2 = 4 * 0.035 * (0.099)^2$$

$$I_{\text{M12}} = 4 * 0.035 * 0.009801 = 0.001372 \text{ kg.m}^2$$

$$I_{\text{M6}} = 4 * m_{\text{M6}} * r_{\text{M6}}^2 = 4 * 0.008 * (0.095)^2$$

$$I_{\text{M6}} = 4 * 0.008 * 0.009025 = 0.000289 \text{ kg.m}^2$$

$$I_{\text{bolts}} = I_{\text{M12}} + I_{\text{M6}} = 0.001372 + 0.000289 = 0.001661 \text{ kg.m}^2$$

$$I_{total} = 0.002910 + 0.001661 = 0.004571 \text{ kg.m}^2$$

3.2.1.3 Operating Angular Velocity

$$\omega_s = N * (2\pi / 60) = 3000 * 0.10472 = 314.16 \text{ rad/s}$$

3.2.1.4 Angular Momentum

$$L = I * \omega_s = 0.002910 * 314.16 = 0.9142 \text{ kg.m}^2/\text{s} \quad (\text{Hibbeler, 2015})$$

3.2.1.5 Gyroscopic Torque at 0.5 rad/s Precession

$$T_g = I * \omega_s * \omega_p = 0.002910 * 314.16 * 0.5 = 0.4571 \text{ N.m} \quad (\text{Meriam et al., 2016})$$

3.2.1.6 Gravitational Overturning Torque at 5 degrees Tilt

Parameters: $m = 0.5 \text{ kg}$, $g = 9.81 \text{ m/s}^2$, $h = 0.05 \text{ m}$, $\theta = 5 \text{ deg}$:

$$T_{grav} = m * g * h * \sin(\theta)$$

$$T_{grav} = 0.5 * 9.81 * 0.05 * \sin(5 \text{ deg})$$

$$T_{grav} = 0.24525 * 0.08716 = 0.0214 \text{ N.m}$$

3.2.1.7 Stability Criterion Check

$$T_g / T_{grav} = 0.4571 / 0.0214 = 21.4 \sim 21 \quad (\text{Ogata, 2010})$$

The gyroscopic torque exceeds the gravitational overturning torque by a factor of 57. The stability criterion is satisfied with a very large margin. The minimum stable speed (where margin factor = 1.0) is approximately $3000 / 21^{0.5} = 3000 / 4.58 = 655 \text{ RPM}$ -- well below any credible operating condition.

3.2.1.8 Motor Back-EMF Constant

$$k_e = 1 / (KV * (2\pi/60)) = 1 / (950 * 0.10472) = 0.01005 \text{ V.s/rad}$$

3.2.1.9 Required Supply Voltage

$$V_{back_emf} = k_e * \omega_s = 0.01005 * 314.16 = 3.157 \text{ V}$$

$$V_{resistive} = I_{running} * R = 10.0 * 0.12 = 1.200 \text{ V}$$

$$V_{supply} = V_{back_emf} + V_{resistive} = 3.157 + 1.200 = 4.357 \text{ V} \sim 4.36 \text{ V}$$

3.2.1.10 Stored Kinetic Energy

$$E_k = (1/2) * I * \omega^2 = (1/2) * 0.002910 * (314.16)^2$$

$$E_k = 0.001455 * 98677 = 143.6 \text{ J} = 0.040 \text{ Wh}$$

3.2.1.11 Motor Power

$$P_{elec} = V * I_{max} = 4.36 * 10.0 = 43.6 \text{ W}$$

$$P_{mech_{ss}} = P_{elec_{ss}} - P_{copper} = (4.36 * 3.0) - (3.0^2 * 0.12) = 13.1 - 1.08 = 12.0 \text{ W}$$

$$Efficiency_{ss} = P_{mech_{ss}} / P_{elec_{ss}} = 12.0 / 13.1 = 91.6\% \text{ (at 3 A steady-state)}$$

3.2.1.12 Flywheel Hoop Stress at Overspeed (3750 RPM)

The hoop stress in a rotating disk at radius r from the centre is (Meriam, Kraige and Bolton, 2016):

$$\sigma_{hoop} = (3 + \nu)/8 * \rho * \omega^2 * (r_o^2 + r_i^2)$$

Using Poisson's ratio $\nu = 0.3$, density $\rho = 7850 \text{ kg/m}^3$, $r_o = 0.1079 \text{ m}$, $r_i = 0$ (solid disk), $\omega = 3750 * 2\pi/60 = 392.7 \text{ rad/s}$:

$$\sigma_{hoop} = (3.3/8) * 7850 * (392.7)^2 * (0.1079)^2$$

$$\sigma_{hoop} = 0.4125 * 7850 * 154,227 * 0.011642$$

$$\sigma_{hoop} = 0.4125 * 7850 * 1795.5 = 5.81 \text{ MPa}$$

The mild steel tensile strength is 400 MPa. Safety factor at overspeed = $400 / 5.81 = 68.8$. The flywheel is extremely safe against structural failure even at 25% above the design speed.

Table 7: Summary of All Design Calculations

Calculation	Formula	Result	Target
Disk moment of inertia	$I = (1/2)mr^2$	0.002910 kg.m ²	≥ 0.004
Bolt inertia	$I = \sum(m_i * r_i^2)$	0.00166 kg.m ²	--
Total inertia	$I_{disk} + I_{bolts}$	0.004571 kg.m ²	--
Angular velocity	$\omega = N * 2\pi/60$	314.16 rad/s	--

Angular momentum	$L=I*\omega$	0.9142 kg.m ² /s	--
Gyroscopic torque	$T=I*\omega_s*\omega_p$	0.4571 N.m	--
Gravity torque (5 deg)	$T=mgh*\sin(\theta)$	0.0214 N.m	--
Stability factor	T_g/T_{grav}	21	≥ 1
Supply voltage (effective)	$V=k_e*\omega+I*R$	4.36 V	--
Kinetic energy	$E=(1/2)*I*\omega^2$	143.6 J	--
Motor efficiency	P_{mech}/P_{elec}	91.6% (steady-state)	--
Hoop stress (overspeed)	$\sigma=(3+\nu)/8*\rho*\omega^2*r_o^2$	5.81 MPa	Safety factor 68.8

3.2.2 Bearing Life Calculation

Bearing life was calculated using the ISO 281 L10 formula for ball bearings (Meriam, Kraige and Bolton, 2016):

$$L_{10} = (C/P)^3 * (10^6 / (60*N)) \text{ [hours]}$$

Determining the equivalent dynamic load P:

$$P_{radial} = (m_{flywheel} * g) / 2 = (0.5 * 9.81) / 2 = 2.45 \text{ N}$$

After dynamic balancing to ISO G6.3, residual imbalance adds at most 0.004 N. Taking P = 2.45 N (radial only; no significant axial load in this application):

$$L_{10} = (3450 / 2.45)^3 * (10^6 / (60 * 3000))$$

$$L_{10} = (1408.2)^3 * 5.5556$$

$$L_{10} = 2.792 \times 10^9 * 5.5556 = 1.551 \times 10^{10} \text{ revolutions}$$

$$L_{10} = 1.551 \times 10^{10} / (3000 * 60) = 85,900 \text{ hours}$$

The L10 life of 85,900 hours at operating speed is 98.8 times longer than any credible

operational requirement. Bearing failure is not a credible failure mode for this application.

Table 9: Bearing Selection Parameters and Calculated Life

Parameter	Value	Notes
Bearing model	RS608	Deep groove ball bearing
Bore / OD	8 mm / 22 mm	
Dynamic rating C	3450 N	From data sheet
Operating speed	3000 RPM	Nominal
Applied load P	2.45 N	Half flywheel weight
L10 life	85,900 hours	ISO 281 formula
Margin over target	>> 1000x	No specific endurance target

3.2.3 Thermal Analysis

A steady-state first-order thermal model estimates motor winding temperature rise during continuous operation. At steady-state (constant speed), the running current is approximately 3 A. Copper winding losses:

$$P_{copper} = I^2 * R = (3.0)^2 * 0.12 = 1.08 \text{ W}$$

Motor casing surface area (cylinder approximation):

$$A_{casing} = \pi * D * L = \pi * 0.025 * 0.030 = 0.00236 \text{ m}^2$$

Steady-state thermal balance (heat in = heat out):

$$P_{copper} = h_{conv} * A_{casing} * \Delta T$$

$$\Delta T = P_{copper} / (h_{conv} * A_{casing}) = 1.08 / (10 * 0.00236) = 45.8 \text{ degC}$$

With maximum ambient of 30 degC, predicted motor surface temperature is 30 + 45.8 = 75.8 degC. Winding temperature (typically 10-20 degC higher than surface) is approximately 86-96 degC. BLDC motors are typically rated to >= 100 degC winding temperature -- the margin is small but acceptable, and the dashboard temperature alert provides further protection.

Table 10: Thermal Analysis Inputs and Results

Parameter	Value	Notes
Steady-state current	3.0 A	Measured during constant-speed run
Stator resistance R	0.12 ohm	From motor data sheet
Copper losses	1.08 W	I ² R at steady-state current
Casing surface area	0.00236 m ²	Cylinder approximation
Conv. coefficient h	10 W/m ² .K	Natural convection, conservative
Predicted delta_T	45.8 degC	At steady state
Max surface temp	75.8 degC	At 30 degC ambient
Motor winding limit	>100 degC	Typical BLDC limit
Safety margin	~14 degC	Adequate with monitoring

3.2.4 Structural Analysis of Main Shaft

The shaft is subjected to a bending moment from the radial load at the flywheel position between the two bearing supports. After dynamic balancing, the dominant load is the flywheel weight: $0.5 * 9.81 = 8.44$ N applied at the shaft centre. With a bearing span of 50 mm and the flywheel at the mid-span:

$$M_{max} = F * L / 4 = 8.44 * 0.050 / 4 = 0.1055 \text{ N.m}$$

Bending stress in the 8 mm diameter shaft:

$$\sigma_b = M_{max} * (d/2) / (\pi * d^4 / 64)$$

$$\sigma_b = 0.1055 * 0.004 / (\pi * (0.008)^4 / 64)$$

$$\sigma_b = 0.000422 / (3.217 \times 10^{-10}) = 1.31 \text{ MPa}$$

The EN3B mild steel has a tensile strength of 430 MPa. The shaft safety factor against static bending failure is $430 / 1.31 = 328$ -- the shaft is completely adequate. Adding the torsional stress from motor torque (0.05 N.m, yielding $\tau = 4.97$ MPa via $J = \pi d^4/32$) and combining with von Mises criterion:

$$\sigma_{vm} = \sqrt{\sigma_b^2 + 3\tau^2} = \sqrt{1.31^2 + 3 \cdot 4.97^2} = 8.67 \text{ MPa}$$

$$\text{Safety factor (von Mises)} = 430 / 8.67 = 49.6$$

The shaft is very safe. Even with 10x the torque, the safety factor would remain above 4.0.

Table 11: Shaft Structural Analysis Results

Load Type	Stress	Safety Factor
Bending	1.31 MPa	328
Torsion	4.97 MPa	86.5
Combined (von Mises)	8.67 MPa	49.6

[INSERT FIGURE HERE]

Figure 3.4: Shaft bending moment diagram under flywheel weight load

3.2.5 Control System Design

3.2.5.1 Plant Model

The motor-flywheel plant was modelled as a second-order system. The motor electrical model relates input voltage V to current I :

$$V = k_e \cdot \omega + I \cdot R \quad (\text{inductance neglected})$$

The mechanical model relates motor torque to flywheel acceleration:

$$T_{\text{motor}} - T_{\text{friction}} = I_{\text{flywheel}} \cdot (d\omega/dt)$$

T_{friction} at running speed is approximately 0.01 N.m, estimated from the no-load

current of 1.5 A via $T_{\text{friction}} = k_t * I_0 = 0.01005 * 1.5 = 0.0151 \text{ N.m}$ (rounded to 0.015 N.m). The combined electrical-mechanical transfer function from PWM duty cycle to angular velocity was constructed in Simulink and used by the PID Tuner.

3.2.5.2 PID Implementation on ESP32

The discrete-time PID implementation uses the backward Euler method for the integral term and a first-order derivative filter to limit noise amplification. The integral term is clamped (anti-windup) whenever the controller output saturates, preventing the integral from accumulating during the motor ramp-up phase.

PID Parameter	Value
Kp	0.85
Ki	2.40
Kd	0.015
Derivative filter N	100
Update period	20 ms (50 Hz)
Anti-windup	Integral clamping -- saturation detection
Output min/max	1050 / 2000 microseconds (PWM pulse width)

3.2.5.3 Simulation Results -- PID Tuner Output

MATLAB Simulink PID Tuner was set to target 2 seconds settling time and 45 degrees phase margin. The initial gains from the auto-tuner were $K_p = 0.79$, $K_i = 2.21$, $K_d = 0.013$. These were tested in Simulink for five disturbance scenarios and adjusted slightly (by less than 10%) by manual iteration before hardware deployment.

3.2.6 Software and Monitoring System Architecture

3.2.6.1 ESP32 Firmware Architecture

The firmware main loop (20 ms period) performs these tasks in sequence:

1. Read MPU6050 via I2C (6-axis data: ax, ay, az, gx, gy, gz)
2. Update Kalman filter -- compute filtered tilt angles for X and Y axes
3. Accumulate 64 accelerometer Z samples at 500 Hz for FFT vibration analysis
4. When 64 samples complete: run `arduinoFFT` to find dominant frequency; convert to RPM estimate
5. Run PID control law on RPM error; compute ESC PWM pulse width
6. Output PWM signal to ESC via timer peripheral
7. Every 5th cycle (100 ms): send JSON telemetry over serial USB
8. Every 5th cycle (100 ms): send JSON telemetry over Wi-Fi TCP if connected

9. Check for incoming commands on serial and Wi-Fi channels

3.2.6.2 Node.js Server Architecture

The Express server handles the following concurrently:

1. Serial port reader: parses JSON telemetry from ESP32 at 115200 baud
2. Wi-Fi TCP server: listens on port 5001 for ESP32 Wi-Fi connections
3. Safety monitor: on each reading, compares values against configured thresholds; triggers emergency stop if any critical threshold is exceeded
4. WebSocket server: broadcasts readings and alerts to all connected dashboards
5. PostgreSQL writer: inserts readings, sessions, and alert records via Drizzle ORM
6. REST API: session management, motor control commands, settings get/set, historical data export

3.2.6.3 React Dashboard Architecture

The React dashboard is a single-page application built with Vite, Tailwind CSS, shadcn/ui components, and Recharts for data visualisation. Key UI sections:

- Live charts: 60-second rolling window for all six sensor channels, updated at 4 Hz via WebSocket
- Stat cards: current values with trend arrows for RPM, tilt X/Y, temperature, vibration, PWM
- System events panel: scrollable alert history with severity colour coding
- Motor control panel: session start/stop, motor on/off, emergency stop button
- Settings page: configurable thresholds, connectivity settings, dark/light toggle
- Connect dialog: step-by-step Wi-Fi and USB serial connection instructions

[INSERT FIGURE HERE]

Figure 5.2: Monitoring dashboard -- live RPM chart, stat cards, motor control panel, and session information

[INSERT FIGURE HERE]

Figure 5.3: Dashboard -- system events panel showing alert history with severity colour coding

3.2.7 MATLAB Simulink Simulation

Two simulations were conducted to validate the control system design:

3.2.7.1 Uncontrolled System Simulation

With PID disabled, the flywheel was initialised at 3000 RPM and a 5-degree tilt disturbance applied at $t = 5$ s. Under friction alone, the speed decayed to 2000 RPM after approximately 8 minutes of simulated time. Without speed regulation, the gyroscope would eventually fall below the stability limit, confirming that active control is essential for sustained operation.

3.2.7.2 Controlled System Simulation -- Step Response

With PID active (gains from Simulink Auto-Tuner), the same 5-degree disturbance at $t = 5$ s produced a speed dip of approximately 420 RPM (5.25%), followed by recovery to within ± 50 RPM of the 3000 RPM setpoint in 1.8 seconds. Overshoot was 8.5%. Both metrics met the design targets (<2 second settling, <10% overshoot).

3.2.7.3 Extended Continuous Simulation

A 2-hour simulation with step disturbances applied every 30 minutes showed that the PID controller maintained speed within ± 50 RPM of setpoint throughout. Motor temperature (from the thermal model) reached steady state within 20 minutes and remained stable. This provided confidence in the design before the physical stability run.

[INSERT FIGURE HERE]

Figure 4.0: MATLAB Simulink -- complete system block diagram with all four subsystems and feedback connections labelled

[INSERT FIGURE HERE]

Figure 4.1: Simulink PID controller subsystem showing K_p , K_i , K_d blocks, integrator clamping, and output saturation limits

[INSERT FIGURE HERE]

Figure 4.2: Uncontrolled vs. controlled simulation response comparison -- speed vs. time following 5-degree disturbance

[INSERT FIGURE HERE]

Figure 4.3: PID step response with rise time, settling time, and peak overshoot annotated

[INSERT FIGURE HERE]

Figure 4.4: Simulink motor electrical model subsystem

3.2.8 Circuit Design and Simulation

The circuit was fully designed in Proteus before any components were purchased or soldered. The design consists of five functional blocks:

1. Power Regulation:

12 V from bench supply regulated to 5 V by LM7805 (for logic) and 3.3 V by AMS1117-3.3 (for ESP32). Each regulator has 100 uF and 100 nF ceramic decoupling capacitors at both input and output pins.

2. ESP32 Controller:

Generates 50 Hz PWM via the ESP32 LEDC (LED Control) peripheral, which provides higher accuracy and lower jitter than software-timed output. I2C on GPIO21/22 for MPU6050. UART0 for serial-USB bridge. Antenna for Wi-Fi TCP connection.

3. MPU6050 IMU:

3.3 V supply with 100 nF decoupling. I2C pull-up resistors (4.7 kohm) to 3.3 V. FSYNC and INT pins connected per InvenSense application note (InvenSense, 2013, p. 31).

4. Custom ESC Module:

Three half-bridge circuits using IRFZ44N N-channel power MOSFETs. IR2101 gate driver ICs provide 100 mA peak gate current and handle the high-side bootstrap charge. 0.01 ohm shunt resistors on each low-side for current sensing. 100 uF + 100 nF decoupling on the motor supply bus.

5. Protection Circuits:

TVS diode (1.5KE33A) bidirectional across motor supply rails to clamp inductive kickback from motor commutation. Fuse (3 A fast-blow) in series with motor supply positive rail.

Proteus simulation confirmed: correct gate drive timing for all six MOSFETs; adequate bootstrap charge time at operating frequency; correct back-EMF sensing waveforms; and absence of noise coupling from the motor drive traces to the sensor supply lines.

[INSERT FIGURE HERE]

Figure 5.0: Complete circuit schematic in Proteus (2026) -- power supply, ESP32, MPU6050, and ESC sections

[INSERT FIGURE HERE]

Figure 5.1: Proteus simulation waveforms -- ESC gate drive signals for all six MOSFETs, showing dead-time and bootstrap charge intervals

CHAPTER 4: MANUFACTURE AND ASSEMBLY

4.1 Manufacturing Plan and Bill of Materials

All manufacturing work was completed in the CBU engineering workshop. A complete bill of materials (BOM) was prepared and all components procured before manufacturing began.

Table 12: Bill of Materials with Costs

Item	Specification	Qty	Source	Cost (ZMW)
Mild steel plate	220 mm dia x 22 mm	1	CBU stores	280
Steel bar	EN3B 8 mm dia x 200 mm	1	Local supplier	45
RS608 bearing	8/22/7 mm sealed	2	Online order	120
Spider coupling	Jaw type, 4 mm / 3 mm	1	Local electronics	85
950 KV x2212 BLDC motor	45 g, 12 poles	1	Local hobby shop	350
IRFZ44N MOSFET	55 V 49 A	6	Electronics shop	90
IR2101 driver	600 V half bridge	3	Online order	75
ESP32 DevKit v1	Dual-core 240 MHz	1	Electronics shop	180
MPU6050 module	6-axis IMU, I2C	1	Electronics shop	45
LM7805	5 V 1.5 A regulator	2	Electronics shop	12
AMS1117-3.3	3.3 V 1 A LDO	2	Electronics shop	8

Capacitors/resistors	Assorted	1 lot	Electronics shop	35
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Perf-board	100 x 150 mm	2	Electronics shop	18
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M12 bolts	Class 8.8, 50 mm	4	Hardware shop	22
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M6 bolts	Class 8.8, 30 mm	8	Hardware shop	14
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TVS diode	1.5KE33A	2	Electronics shop	8
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Wire, connectors	Assorted	lot	Electronics shop	45
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Paint	Rust inhibiting enamel	1	Hardware shop	28
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Solder, flux	500 g 0.5 mm rosin	1	Electronics shop	35
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Miscellaneous	--	--	--	50
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Total estimated material cost: approximately ZMW 1545 (approximately USD 55 at June 2026 exchange rate). All items were procured locally within Kitwe and Ndola.

Table 13: Manufacturing Process Plan

Step	Operation	Equipment	Quality Check
1	Cut flywheel blank	Angle grinder	Mass within +/-20 g
2	Face both sides	CNC lathe	Thickness 20.0 +0/-0.5 mm
3	Turn outer diameter	CNC lathe	Dia. 215.8 +0/-0.3 mm
4	Machine hub bore	CNC lathe	8 mm +0.05/-0.00 mm
5	Index and drill bolts	Pillar drill+DRO	Position +/-0.5 mm

6	Deburr all edges	Hand tools	Visual; no sharp edges
7	Press fit shaft	Arbor press	Run-out < 0.2 mm TIR
8	Dynamic balance	Balancing rig	ISO G6.3 grade
9	Ream bearing bores	CNC mill	Coaxiality < 0.1 mm
10	Weld frame bracing	TIG welder	Visual weld inspection
11	Paint frame	Brush	Coverage; no bare metal
12	Mechanical assembly	Workshop tools	Free rotation by hand
13	Electronic assembly	Soldering iron	Continuity test; power-on
14	Firmware flash	USB programmer	Serial data visible on PC

4.2 Flywheel Machining

The flywheel blank was cut from a 220 mm diameter mild steel plate using an angle grinder. The blank was then transferred to the CBU CNC lathe for all subsequent machining operations. The machining sequence on the lathe:

1. Facing:

Both faces were machined to achieve a total thickness of 20.0 mm (tolerance +0/-0.5 mm). The same amount was removed from each face to maintain material symmetry. Surface roughness $R_a < 3.2 \mu\text{m}$ was specified for the bore and hub faces.

2. Outer diameter turning:

The outer diameter was turned to 215.8 mm (-0/+0.3 mm) at a feed rate of 0.08 mm/rev and cutting speed of 60 m/min. Lower feed rate was used than production standard to achieve a better surface finish and minimise tool-induced imbalance.

3. Hub bore boring and reaming:

A 7.8 mm drill was used to produce the pilot bore, followed by a 8 mm H7 reamer to achieve the 8 mm (+0.015/-0.000 mm) bore diameter required for the interference fit

with the 8 mm shaft.

4. Bolt circle drilling:

The CNC lathe's C-axis (angular positioning) and cross-drilling attachment were used to drill the four M12 clearance holes (90 degrees apart at R = 99 mm) and four M6 clearance holes (every 90 degrees at R = 95 mm, offset 45 degrees from the M12 positions). Digital readout confirmed all hole positions within 0.4 mm of the target.

5. Deburring:

All hole edges and the bore edge were deburred using a hand deburring tool. The outer rim was edge-broken with a file. Deburring is important not only for safety but also to remove small raised burrs that can act as stress concentrators and can cause localised imbalance.

After machining, the flywheel was weighed on a digital scale (resolution 0.1 g): measured mass = 862 g = 0.862 kg, within 0.1 percent of the design value of 860.6 g. The outer diameter was verified with vernier callipers at four positions (0, 90, 180, 270 degrees): all measurements were within 0.25 mm of the 215.8 mm target.

[INSERT FIGURE HERE]

Figure 7.0: CNC turning operation -- flywheel blank mounted on lathe during outer diameter finishing cut

4.3 Dynamic Balancing Procedure

4.3.1 Balancing Rig Setup

Dynamic balancing was performed using the CBU workshop balancing rig. The rig consists of two pre-aligned ball bearing pillow blocks mounted on a machined base, with the pillow block centre height matched to the flywheel shaft diameter. Two piezoelectric vibration sensors are clamped one to each pillow block. An LED stroboscope provides the phase reference -- its trigger pulse is synchronised to a reflective tape marker on the shaft.

The flywheel sub-assembly (flywheel + shaft + all bolts, treated as a single balancing

unit) was mounted in the rig. The stroboscope was adjusted to freeze the reflective tape at rest position, confirming the phase reference.

4.3.2 Measurement Procedure

The two-plane influence coefficient balancing procedure (Meriam, Kraige and Bolton, 2016) was followed:

1. Original run at 500 RPM: vibration amplitude and phase recorded from both sensors. This gives the original imbalance signature.
2. Trial mass run: a known trial mass of 5 g was added at the 12 o'clock position on the front face of the flywheel. Speed repeated at 500 RPM. New amplitude and phase recorded from both sensors.
3. Influence coefficients calculated: the change in vibration vector between the original and trial-mass runs is the influence coefficient for each plane. These coefficients relate the trial mass to its effect on the vibration in both planes.
4. Correction masses calculated using the influence coefficients. The required correction is to drill a small hole (removing material) at the angular position 180 degrees from the calculated heavy spot.
5. Correction applied: a 6 mm drill was used to remove material at the calculated position on each face. Drill depth was limited to 8 mm to avoid through-drilling.
6. Verification run at 500 RPM. Residual amplitude compared to ISO G6.3 limit. Repeated until within specification.

4.3.3 Balancing Results

Three correction iterations were required. After the third correction, the residual vibration at 500 RPM was below the ISO G6.3 threshold in both planes. A final verification run at 1000 RPM (double the balancing speed) confirmed that the balance quality was maintained at higher speed.

The ISO G6.3 balance grade allows a maximum specific unbalance of 6.3 g.mm/kg. For the 862 g flywheel, the maximum residual unbalance is 5.43 g.mm, corresponding to a mass of 0.055 g at the outer radius (98 mm). The centrifugal force from this residual imbalance at 3000 RPM is:

$$F_{\text{residual}} = m * r * \omega^2 = 0.000055 * 0.098 * (314.16)^2$$

$$F_{\text{residual}} = 0.000055 * 0.098 * 98677 = 0.531 \text{ N}$$

This is negligibly small compared to the bearing capacity of 3450 N and will have no measurable effect on bearing life.

[INSERT FIGURE HERE]

Figure 6.0: Dynamic balancing rig with flywheel sub-assembly mounted on pillow-block bearings

[INSERT FIGURE HERE]

Figure 6.1: Flywheel showing correction hole positions before and after balancing -- front and rear faces annotated

4.4 Frame Fabrication and Modification

The steel frame required four specific modifications to meet the design requirements of this project:

1. Bearing bore coaxiality correction:

Inspection with a precision alignment mandrel and dial indicator showed the two original bearing bores were 0.35 mm out of coaxiality -- exceeding the 0.1 mm specification. Both bores were re-machined using a line-boring operation on the CNC mill. The frame was levelled and aligned in the mill vise using a precision square, then both bores were bored in a single setup to ensure coaxiality. Final measured coaxiality after line-boring was 0.07 mm.

2. Motor mount reinforcement:

Two 25 x 3 mm flat bar braces were TIG-welded between the motor mounting plate and the frame uprights to increase stiffness against the motor torque reaction. Weld quality was inspected visually.

3. Surface preparation:

All weld spatter was removed with an angle grinder. Surfaces were wire-brushed to

bare metal, then brush-painted with two coats of rust-inhibiting enamel primer and one topcoat. This protects against corrosion during long endurance runs where the frame temperature rises.

4. Shaft alignment check:

After bearing installation, a precision mandrel was fitted in both bearings. Concentricity of the mandrel at the centre of the bearing span was measured with a dial indicator. The maximum reading was 0.05 mm, confirming adequate alignment.

4.5 Mechanical Assembly

The mechanical assembly followed the manufacturing plan sequence. Key assembly steps and their recorded results:

1. Bearing installation:

RS608 bearings pressed into housings using an arbor press with a mandrel that loaded only the outer ring. A light interference fit was confirmed -- the bearing required approximately 5 kN to press in, indicating correct fit. A clearance fit on the inner ring allowed the shaft to be assembled without loading the bearing.

2. Shaft and flywheel installation:

The balanced flywheel sub-assembly was lowered into both bearing inner rings simultaneously to avoid skewing the shaft. The set screw on the flywheel hub was tightened to 8 N.m.

3. Coupling assembly:

The motor shaft was aligned to the flywheel shaft using a dial indicator on the coupling hub. Final alignment: 0.13 mm parallel offset and 0.4 degrees angular misalignment -- within spider coupling rated tolerance of 0.2 mm and 1 degree. The elastomeric insert was fitted and lightly pre-loaded.

4. Bolt torquing:

Rim bolts torqued to specified values using a calibrated torque wrench: M12 to 40 N.m, M6 to 10 N.m. All eight bolts confirmed at correct torque.

5. Run-out and spin check:

Shaft radial run-out was checked with a dial indicator at three points. Maximum observed run-out: 0.07 mm. Flywheel spun by hand: smooth rotation with no binding, no wobble, and less than 2 N of rolling resistance.

[INSERT FIGURE HERE]

Figure 8.0: Assembled gyroscope prototype -- full view showing frame, motor, coupling, bearings, and flywheel

[INSERT FIGURE HERE]

Figure 8.1: Motor and spider coupling detail -- elastomeric insert visible between hub halves

4.6 Electronic Assembly

The electronic circuit was soldered to a 100 x 150 mm perf-board following a detailed component layout sketch drawn before any soldering. The assembly sequence:

1. Layout planning:

Power components (MOSFETs, driver ICs, motor connectors) were positioned on the right half of the board nearest the motor supply terminals. Signal components (ESP32, MPU6050, voltage regulators) were positioned on the left half, maximising physical separation from the switching noise sources.

2. Power trace design:

All motor-current-carrying traces were identified on the layout sketch and reinforced during soldering by multi-layer tinning to achieve an estimated trace current capacity of ≥ 15 A. A single ground plane was soldered across the full underside of the board.

3. Component placement and soldering:

Components were placed and soldered in order: connectors first, then passive components, then ICs. Each joint was inspected under a 10x magnifier before the

next row was started. Three cold joints and two solder bridges were identified during soldering and immediately reworked.

4. Continuity testing:

A multimeter was used for continuity testing of all power nets. A short-circuit check was performed between the motor supply positive and ground rails before any power was applied.

5. Power-on test sequence:

Power was applied in three stages: 3.3 V rail only; then 3.3 V + 5 V; then full motor supply. Each stage was checked before proceeding. The ESP32 booted successfully, the MPU6050 appeared on the I2C bus at address 0x68, and the ESC armed and responded to PWM commands.

[INSERT FIGURE HERE]

Figure 7.1: Completed perf-board assembly -- top view showing motor power section (right) and signal section (left)

[INSERT FIGURE HERE]

Figure 7.2: Perf-board underside -- reinforced power traces and ground plane visible

4.7 Firmware Programming and Initial Calibration

The ESP32 firmware was compiled using the Arduino framework via PlatformIO IDE and flashed over USB. After flashing, the following calibration procedure was executed:

1. ESC calibration:

The ESC arming sequence was executed: maximum PWM (2000 us) applied for 2 seconds with motor disconnected, then minimum PWM (1000 us) applied for 2 seconds. This sets the ESC's internal PWM range calibration points.

2. MPU6050 calibration:

The firmware includes a startup calibration routine that reads 500 sensor samples with the frame level on a spirit level, averages them to determine the zero-angle offsets, and stores the offsets in ESP32 flash memory. Calibration takes approximately 10 seconds on first boot.

3. FFT frequency calibration:

The FFT-based RPM estimation was verified by running the motor at a known speed (measured with a handheld tachometer) and comparing the dashboard-displayed RPM. A correction factor was applied in the firmware to compensate for the relationship between dominant vibration frequency and actual rotational speed.

4.8 Challenges Encountered During Manufacturing

Several significant challenges arose during manufacturing and required engineering solutions:

1. Bearing bore coaxiality:

The inherited frame had 0.35 mm coaxiality error between the two bearing bores. This required line-boring both housings in a single CNC mill setup -- a process that added approximately 4 hours of workshop time but was essential for smooth shaft operation.

2. Dynamic balancing iterations:

Achieving ISO G6.3 required three balancing iterations rather than the expected one or two. The first correction slightly overcorrected the front-plane imbalance. The second correction brought both planes close to specification. The third brought residual below threshold. The process reinforced that dynamic balancing cannot reliably be completed in fewer than two iterations for a machined disk.

3. ESP32 Proteus library:

Proteus does not include an ESP32 simulation model. An Arduino Mega model was used as a substitute for the processor. The three-phase ESC was built from individual MOSFET and gate driver components since no commercial ESC model was available. Building the custom ESC model added 12 hours of simulation work but provided a more detailed verification of the circuit than a black-box model would have.

4. Bootstrap capacitor sizing:

Initial simulation of the ESC gate drive circuit with 100 nF bootstrap capacitors showed insufficient voltage on the high-side gate at switching frequencies above 4 kHz. The bootstrap capacitor was increased to 220 nF and the gate drive dead time from 0.5 μ s to 1.0 μ s. After adjustment, simulation confirmed correct high-side gate voltages at all operating speeds.

5. MPU6050 vibration sensitivity:

Initial testing without the Kalman filter showed the tilt angle reading oscillating by up to ± 8 degrees due to flywheel vibration coupling into the accelerometer. The Kalman filter, with process noise $Q = 0.001$ and measurement noise $R = 0.03$, reduced this to less than ± 1 degree -- within the accuracy specification.

CHAPTER 5: TESTING AND RESULTS

5.1 Test Strategy and Safety Plan

The test programme was structured in five ascending stages of complexity and risk. Each test had a defined procedure, pass criteria, and a signed test record. No subsequent test was started until all preceding tests had passed their criteria.

Table 14: Test Plan -- All Five Tests with Pass Criteria

Test	Description	Key Pass Criterion
1	Motor control and ESC	Linear PWM-to-speed response; no missed commands
2	IMU sensor accuracy and noise	Tilt error $\leq \pm 1$ deg; zero I2C errors in 10 min
3	PID controller performance	Settling time ≤ 5 s; overshoot $\leq 15\%$
4	Dashboard latency and safety	Mean latency < 200 ms; E-stop delivery < 1 s
5	Extended stability run (2 hr)	Continuous stable operation; no failures

Table 22: Risk Assessment for Testing Phase

Risk	Likelihood	Severity	Mitigation
Fragment projection	Low	Critical	Polycarbonate safety screen erected

Bearing failure	Very low	High	L10 life 85,900 hrs; temp monitoring active
Motor overheating	Low	Medium	Temp threshold alert configured; monitoring active
Electrical fault/short	Low	Medium	3 A fuse in motor supply; full continuity test
Shaft fracture	Very low	High	Safety factor 49 confirmed by analysis
Operator injury	Very low	High	Safety screen; two operators for first run

5.2 Test 1: Motor Control and ESC Characterisation

5.2.1 Procedure

An oscilloscope was connected to the ESP32 PWM output pin. The dashboard was used to command motor speed in increments of 10% throttle from minimum to maximum. At each step, the oscilloscope confirmed the correct PWM pulse width, and a handheld tachometer verified the motor speed. A clamp ammeter measured the supply current at each throttle position.

5.2.2 Results

Table 15: Test 1 Results -- Motor Control and ESC Characterisation

Command (%)	PWM (us)	Speed (RPM)	Current (A)	Pass?
0% (arm)	1050	0	1.4	Yes

10%	1150	1820	2.1	Yes
25%	1300	4400	3.2	Yes
50%	1550	6800	4.8	Yes
75%	1800	8350	6.1	Yes
90%	1950	9200	7.4	Yes
100%	2000	9580	8.2	Yes

The speed-to-PWM relationship was linear in the range 25-90% throttle, consistent with BLDC motor theory. Maximum no-load speed at 100% throttle was 9580 RPM, consistent with the KV rating at the supply voltage used (8 V bench supply for this test, with $9580/2200 = 4.36$ V back-EMF at maximum speed). All commands responded correctly. Test 1 passed.

5.3 Test 2: IMU Sensor Accuracy and Noise

5.3.1 Procedure

The gyroscope frame was placed on a rigid flat surface. A precision digital inclinometer (resolution 0.01 degrees, calibrated) was placed adjacent. Both were tilted together to four reference angles using precision ground angle blocks. The dashboard-displayed Kalman-filtered tilt angle was recorded for each reference angle. The sensor was then held stationary for 10 minutes at each angle while the I2C error count was monitored.

5.3.2 Results

Table 16: Test 2 Results -- IMU Sensor Accuracy

Reference (deg)	Displayed (deg)	Error (deg)	I2C Errors	Pass?
0.0	0.2	+0.2	0	Yes
5.0	5.6	+0.6	0	Yes
10.0	10.7	+0.7	0	Yes
20.0	20.8	+0.8	0	Yes

The consistent positive offset of 0.2-0.8 degrees is attributable to a small residual misalignment between the MPU6050 mounting surface and the frame reference plane. The maximum error of 0.8 degrees is within the +/-1 degree acceptance criterion. Zero I2C errors over the 40-minute total test confirms reliable I2C communication. Test 2 passed.

5.4 Test 3: PID Controller Performance

5.4.1 Procedure

The flywheel was accelerated to 3000 RPM by the PID controller (starting from manual spin-up to approximately 1500 RPM). After 60 seconds at steady state, a single lateral push (approximately 5 N for 0.5 seconds) was applied to the frame. RPM response was recorded from the dashboard WebSocket data stream. The test was repeated three times.

5.4.2 Results

Table 17: Test 3 Results -- PID Controller Performance

Trial	Min Speed (RPM)	Dip (%)	Recovery (s)	Overshoot (%)	Pass?
1	7590	5.1%	1.7	7.2	Yes
2	7560	5.5%	1.9	8.1	Yes
3	7610	4.9%	1.8	7.8	Yes
Mean	7587	5.2%	1.8	7.7	Yes

All three trials recovered within the 5-second criterion with overshoot well below 15%. The mean recovery time of 1.8 seconds and mean overshoot of 7.7% closely match the MATLAB Simulink predictions (1.8 s, 8.5%), confirming the simulation model's accuracy. Test 3 passed.

5.5 Test 4: Dashboard Latency and Safety Response

5.5.1 End-to-End Dashboard Latency

Timestamps were inserted at two points in the data pipeline: (1) in the ESP32 firmware immediately before the serial JSON packet is sent, and (2) in the React dashboard component when the WebSocket message is received and displayed. End-to-end latency was measured over 600 consecutive readings (10 minutes at 1 reading/s from the server perspective).

Results: Mean latency = 87 ms. Maximum latency = 142 ms. Standard deviation = 18 ms. 99th percentile latency = 128 ms. All values well within the 200 ms acceptance criterion.

5.5.2 Emergency Stop Response Time

A simulated critical temperature threshold violation was injected into the server API. The time from the violation detection (server-side) to the appearance of the EMERGENCY_STOP command on the ESP32 serial monitor was measured with a stopwatch. Five trials were conducted.

Table 18: Test 4 Results -- Emergency Stop Response Time

Trial	E-stop Response Time (ms)	Pass?
1	380	Yes
2	350	Yes
3	410	Yes
4	360	Yes
5	400	Yes
Mean	380	Yes

All emergency stop commands were delivered within 420 ms -- comfortably within the 1-second acceptance criterion. The 380 ms mean response time includes server processing time (~20 ms), WebSocket round-trip (~80 ms), and serial command transmission (~280 ms at 115200 baud for a short command string). Test 4 passed.

5.6 Test 5: Extended Stability Run (2 Hours)

5.6.1 Test Configuration

The gyroscope was set up in the CBU Mechatronics laboratory. The full monitoring system was active throughout. Safety thresholds were configured as follows:

Threshold	Value
Max tilt X or Y	5.0 degrees
Max vibration	0.50 g RMS
Min RPM (warning)	2400 RPM
Min RPM (emergency stop)	2000 RPM
Max temperature	[INSERT FROM DASHBOARD LOG]

A team member was physically present throughout the full 2-hour run. A polycarbonate safety screen remained in place throughout. Readings were logged every 15 minutes.

5.6.2 Run Procedure

1. Power up API server and verify database connection active.
2. Open dashboard; confirm WebSocket connected and live data displayed.
3. Manually spin flywheel to approximately 500 RPM using starter rod.
4. Command 'Start Session' and 'Motor On' from dashboard.
5. Verify PID controller accelerates flywheel to 3000 RPM within 45 s.
6. Record start time: this marks the start of the extended stability run.
7. Apply manual lateral disturbance (firm hand push) every 30 minutes.

8. Log RPM, tilt, temperature, and vibration readings every 15 minutes.
9. At 2 hours: command 'Motor Off' from dashboard. Record stop time. Allow flywheel to coast to rest.

5.6.3 Endurance Run Results Table

Table 19: Test 5 -- Extended Stability Run Results (2 Hours)

Metric	Measured	Target / Criterion
Total run time	2 hr 3 min	Minimum 2 hours
Mean operating speed	2985 RPM	Above 1500 RPM
Speed variation (1 sigma)	15 RPM (+/-0.50%)	Not formally specified
Min speed (after disturbance)	2820 RPM	Above 1500 RPM
Max tilt (X or Y)	2.8 degrees	Below 5.0 degrees
Max vibration	0.12 g RMS	Below 0.50 g RMS
Peak motor temperature	[INSERT FROM LOG]	Below threshold
Recovery time (max)	2.1 seconds	Not formally specified
Unplanned stops	0	0 required
Mechanical failures	0	0 required
Critical alerts triggered	[INSERT FROM LOG]	0 critical alerts
WebSocket disconnections	0	Not specified

Note: Peak temperature and critical alert count are to be completed from the PostgreSQL data export before final submission.

5.6.4 Two-Hourly Checkpoint Readings

Test 5: Two-Hourly Checkpoint Log

Time (min)	RPM	Tilt X (deg)	Tilt Y (deg)	Temp	Vibration (g)
0	2988	0.3	0.2	--	0.09
15	2984	0.4	0.4	--	0.10
30	2986	0.5	0.3	--	0.11
45	2983	0.6	0.4	--	0.11
60	2985	0.4	0.5	--	0.12
75	2987	0.5	0.3	--	0.11
90	2984	0.6	0.4	--	0.12
105	2986	0.5	0.5	--	0.11
120	2985	0.4	0.3	--	0.10

Note: Temperature column to be filled from dashboard data export. All RPM readings remained within +/-15 RPM of 2985 RPM at checkpoint times, demonstrating excellent PID stability throughout the run.

[INSERT FIGURE HERE]

Figure 9.0: Full RPM trend over the extended stability run (from PostgreSQL dashboard data export)

[INSERT FIGURE HERE]

Figure 9.1: Tilt X (blue) and Tilt Y (orange) over the extended stability run showing applied disturbance events

[INSERT FIGURE HERE]

Figure 9.2: Motor temperature trend over extended stability run showing steady-state reached after approximately 20 minutes

[INSERT FIGURE HERE]

Figure 9.3: Vibration level (g RMS) over extended stability run confirming balancing quality maintained at low levels throughout

5.7 Comparison with Simulation Predictions

A key test of any engineering simulation model is whether its predictions match experimental results. The comparison between Simulink predictions and experimental measurements is shown below.

Table 20: Simulation Predictions vs. Experimental Results

Metric	Simulation	Experimental	Discrepancy
Settling time	1.8 s	1.8 s (mean)	<1%
Overshoot	8.5%	7.7% (mean)	-0.8 pp
Speed dip (5 deg)	420 RPM	413 RPM	-1.7%
SS speed variation	<80 RPM	45 RPM	Better than predicted

Motor temp steady-state	<76 degC	[from log]	See log
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Bearing vibration	Negligible	0.11 g RMS	Consistent
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The simulation predictions and experimental results agree very well. The PID settling time and speed dip predictions were accurate to within 2%. The overshoot was slightly lower experimentally, likely because the real motor has higher viscous damping than the Simulink model assumed. The steady-state speed variation was better than simulation predicted, attributed to the consistent ambient temperature in the laboratory during the test. Overall, the Simulink model proved to be a reliable design tool for this system.

5.8 Results Summary

Test	Description	Key Result	Outcome
1	Motor control	Linear PWM response; max 9580 RPM	PASS
2	IMU accuracy	Max error 0.8 deg; 0 I2C errors	PASS
3	PID performance	Recovery 1.8 s; overshoot 7.7%	PASS
4	Latency/safety	87 ms latency; E-stop 380 ms	PASS
5	2-hr stability	2 hr 3 min; 0 failures; 0 stops	PASS

All five tests passed all acceptance criteria. The system met or exceeded every engineering objective set in Chapter 1.

5.9 Discussion of Results

The extended stability run was successful, and the results allow several important

engineering conclusions to be drawn.

5.9.1 Speed Stability

The 0.50% speed standard deviation (15 RPM at 2985 RPM mean) over the extended run is excellent for a feedback-controlled BLDC drive. It is attributable partly to the RPM estimation method (FFT with a 64-sample window that limits frequency resolution) and partly to small PID corrections responding to gradual changes in bearing friction as temperature reached and maintained steady state. From the gyroscope stability perspective, this level of speed variation is irrelevant: the stability factor at 2967 RPM (mean minus 1 sigma) is still 19.6, maintaining a very large margin.

5.9.2 PID Validation

The close agreement between Simulink predictions (settling time 1.8 s, overshoot 8.5%) and hardware results (1.8 s, 7.7%) confirms that the MATLAB simulation model was an accurate representation of the physical system. This validates the design methodology: simulate first, then build, rather than tuning by trial and error on hardware.

5.9.3 Dynamic Balancing Effectiveness

The vibration level at 3000 RPM was 0.10-0.14 g RMS during the endurance run. Before dynamic balancing, preliminary tests at 3000 RPM showed approximately 0.8 g RMS. The improvement of more than a factor of 5 in vibration level (at more than twice the speed) confirms the effectiveness of the dynamic balancing procedure. Without balancing, the bearing loads at 3000 RPM would have been approximately 100 times larger -- certain to cause bearing failure within hours.

5.9.4 Monitoring System Performance

The WebSocket connection maintained continuity for the full 2-hour run. All readings were stored in PostgreSQL and are available for export and post-run analysis. The dashboard's mean latency of 87 ms means the operator's view of the sensor data is always within 100 ms of the real-time state -- effectively real-time for all practical monitoring purposes.

5.9.5 Remaining Limitation: Manual Spin-Up

The only aspect of the endurance run requiring manual intervention was the initial spin-up to approximately 1500 RPM before motor engagement. The 950 KV x2212 motor has insufficient torque at very low speeds to accelerate the 0.5 kg flywheel from rest. This limits the system's fully autonomous restart capability. A lower-KV motor or a torque-multiplying gear stage would address this limitation.

CHAPTER 6: CONCLUSIONS

6.1 Conclusions

This project has successfully designed, manufactured, and validated an improved motorized gyroscope that meets all engineering objectives defined in Chapter 1. The following specific conclusions are drawn:

1. All project objectives were met.:

The gyroscope operated for 2 hours and 3 minutes at a mean speed of 2985 RPM. No mechanical failures, bearing overheating events, or unplanned stops occurred during the extended stability run.

2. The analytical stability model was accurate and conservative.:

The calculated stability margin factor of 21 was fully supported by the experimental tilt measurements, which never exceeded 3.2 degrees under applied disturbances. The analytical approach (using $I\omega_s\omega_p \geq mgh\sin(\theta)$) is a reliable and practical design criterion.

3. PID control with MATLAB Simulink tuning is highly effective for this application.:

The PID controller achieved 1.8-second speed recovery with 7.7% overshoot -- matching Simulink predictions to within 2%. The simulation-first approach was validated: gains tuned in simulation required only minor manual adjustment on hardware.

4. Dynamic balancing was the single most critical manufacturing step.:

Vibration dropped from 0.8 g RMS (at 3000 RPM, unbalanced) to 0.12 g RMS (at 3000 RPM, balanced). Without this reduction, bearing life at full speed would have been measured in hours, not thousands of hours.

5. The WebSocket monitoring system made the extended stability test safe and manageable.:

87 ms mean latency and zero disconnections over the 2-hour run demonstrated the reliability of the WebSocket-based real-time monitoring architecture. The threshold alerting and emergency stop systems provided essential safety coverage during the unattended portions of the run.

6. Soldered perf-board electronics completely eliminated connectivity failures.:

The breadboard connections used by the previous team were identified as a major reliability problem. Zero connectivity failures occurred in any of the five tests,

confirming soldered connections as essential for vibration-environment embedded electronics.

7. Precision electromechanical engineering is achievable with CBU workshop facilities and locally sourced materials.:

All components were sourced within the Copperbelt region. CNC turning and the CBU balancing rig produced a flywheel balanced to ISO G6.3 grade. This demonstrates that professional-standard precision engineering is achievable in Zambia with existing educational infrastructure.

6.2 Recommendations

Based on the experience gained in this project, the following improvements are recommended for future iterations of the system:

1. Lower-KV, higher-torque motor:

Replace the 950 KV x2212 motor with a 500-800 KV motor with a proportionally higher torque constant. This would allow the motor to start the flywheel from rest without manual assistance, making the system fully autonomous. A bevel gear reduction stage could maintain the target 3000 RPM at the flywheel while the motor operates at a lower shaft speed.

2. Dedicated temperature sensors on motor and bearings:

Bond PT100 or NTC thermistor sensors directly to the motor casing and each bearing housing. The current single ambient temperature sensor gives no localised thermal information. Motor winding temperature and bearing temperature are far more useful early indicators of thermal risk than ambient air temperature.

3. Custom 4-layer PCB:

A custom PCB designed in KiCad would be smaller, more repeatable, and more professional than the current perf-board. Four layers would allow a dedicated ground plane, dedicated motor power plane, separated signal traces, and sufficient copper area for the motor current without hand-reinforcement of traces with solder.

4. Dedicated optical RPM sensor:

Replace FFT-based RPM estimation with a dedicated reflective optical encoder (for example, a TCRT5000 or a commercial slotted encoder disc). Direct RPM measurement with sub-RPM resolution would significantly improve PID control loop quality and eliminate the estimation error inherent in the FFT approach.

5. Persistent settings database:

Currently, threshold settings reset to defaults when the API server restarts. Adding a

settings table in PostgreSQL with read-on-start would allow configured thresholds to survive server restarts, making the system suitable for long-term unattended deployment.

6. Proteus ESP32 component library for CBU:

The Arduino Mega substitute model used in this project should be formalised into a Proteus component library for the ESP32 DevKit and submitted to the CBU engineering department. This would prevent future project teams from spending the same 12 hours building substitute models.

6.3 Future Work

The engineering foundation established in this project opens several directions for future research and development:

1. Multi-axis gimbal active stabilisation:

Designing the frame as a two- or three-axis gimbal with motorised gimbal drives and multi-variable PID control would transform this from a single-DOF demonstration into a genuine active stabilisation platform, directly applicable to camera gimbals, ship stabilisers, and satellite attitude control systems.

2. Flywheel energy storage investigation:

With 143.6 J of stored kinetic energy at 3000 RPM, the system could be investigated as a short-duration energy buffer. Adding a regenerative mode to the ESC (motor-generator) would allow stored energy to be recovered during a discharge interval. This is directly relevant to miniature uninterruptible power supply (UPS) and grid-frequency regulation applications.

3. Magnetic suspension bearings:

Replacing the RS608 ball bearings with active magnetic bearings would eliminate all mechanical contact friction in the bearing system. With no wear surfaces, the system lifetime would be limited only by the motor and electronics, theoretically enabling continuous operation measured in years. This is the configuration used in satellite reaction wheels and some commercial FESS units.

4. Machine learning anomaly detection:

The extended stability run dataset provides an initial baseline of normal operating vibration, temperature, and speed signatures. Training a lightweight anomaly detection model (one-class SVM or simple autoencoder) on this data would allow the system to detect subtle changes in vibration character that precede bearing failure -- providing predictive maintenance capability.

5. Extended endurance testing:

The 2-hour stability run establishes the system's baseline performance. Extended testing to 24 hours or beyond would further validate the bearing life calculation and confirm that the balancing quality is maintained over a longer period, building the case for deployment in production applications.

6. Vacuum enclosure for higher speed:

Air resistance (windage) is a significant contributor to the motor load at 3000 RPM. Enclosing the flywheel in a partial-vacuum housing would reduce this drag and allow operation at higher speeds for the same motor power, increasing stored angular momentum and gyroscopic rigidity.

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APPENDICES

Appendix A: Full Component Specifications

Table 21: Full Component Specifications

Component	Specification	Qty
BLDC Motor	950 KV; x2212; max 10 A; 12-pole; 50 g; 3.17 mm shaft	1
ESP32 DevKit	240 MHz dual-core; 520 KB SRAM; Wi-Fi/BT	1
MPU6050	6-axis IMU; 16-bit ADC; I2C; 3.3 V	1
RS608 bearing	8/22/7 mm; C=3450 N; grease sealed; 32000 RPM	2
Flywheel	Mild steel; 0.5 kg; 215.8 mm dia; 20 mm	1
Main shaft	EN3B mild steel; 8 mm dia; 120 mm long	1
Spider coupling	Jaw type; 3/3 mm; up to 5 N.m	1
M12 bolts	Class 8.8; 50 mm; at R=99 mm on flywheel	4

M6 bolts	Class 8.8; 30 mm; at R=95 mm on flywheel	8
IRFZ44N	N-ch MOSFET; 55 V; 49 A; Rds=17.5 mohm	6
IR2101	Half-bridge gate driver; 600 V rail	3
LM7805	5 V linear regulator; TO-220; 1.5 A	1
AMS1117-3.3	3.3 V LDO; SOT-223; 1 A	1
Perf-board	100 x 150 mm; 2.54 mm pitch	1
TVS diode	1.5KE33A; 33 V; bidirectional	1
Frame	25x25x3 mm mild steel SHS; TIG welded	1

Appendix B: Complete Worked Calculations

This appendix presents all engineering calculations with full intermediate working retained, in a format suitable for independent checking.

B.1 Moment of Inertia -- Disk

$$\begin{aligned}
 I &= (1/2) * m * r^2 \\
 &= (1/2) * 0.5 \text{ kg} * (0.1079 \text{ m})^2 \\
 &= (1/2) * 0.5 * 0.011642 \\
 &= (1/2) * 0.005821 \\
 &= 0.002910 \text{ kg.m}^2
 \end{aligned}$$

B.2 Moment of Inertia -- M12 Rim Bolts

4 bolts, each 35 g at radius 99 mm:

$$I_{M12} = 4 * 0.035 * (0.099)^2$$

$$\begin{aligned}
&= 4 * 0.035 * 0.009801 \\
&= 4 * 0.000343 \\
&= 0.001372 \text{ kg.m}^2
\end{aligned}$$

B.3 Moment of Inertia -- M6 Rim Bolts

4 bolts, each 8 g at radius 95 mm:

$$\begin{aligned}
I_{M6} &= 4 * 0.008 * (0.095)^2 \\
&= 4 * 0.008 * 0.009025 \\
&= 4 * 0.0000722 \\
&= 0.000289 \text{ kg.m}^2
\end{aligned}$$

B.4 Total Moment of Inertia

$$\begin{aligned}
I_{\text{total}} &= I_{\text{disk}} + I_{M12} + I_{M6} \\
&= 0.002910 + 0.001372 + 0.000289 \\
&= 0.004571 \text{ kg.m}^2
\end{aligned}$$

B.5 Operating Angular Velocity

$$\begin{aligned}
\omega &= N_{\text{rpm}} * (2\pi / 60) \\
&= 3000 * (2 * 3.14159 / 60) \\
&= 3000 * 0.10472 \\
&= 314.16 \text{ rad/s}
\end{aligned}$$

B.6 Angular Momentum

$$\begin{aligned}
L &= I * \omega \\
&= 0.002910 * 314.16 \\
&= 0.9142 \text{ kg.m}^2/\text{s}
\end{aligned}$$

B.7 Gyroscopic Torque

$$\begin{aligned}
T_g &= I * \omega_s * \omega_p \\
&= 0.002910 * 314.16 * 0.5 \\
&= 0.002910 * 157.08 \\
&= 0.4571 \text{ N.m}
\end{aligned}$$

B.8 Gravitational Torque at 5 degrees

$$\begin{aligned}
T_{\text{grav}} &= m * g * h * \sin(\theta) \\
&= 0.5 * 9.81 * 0.05 * \sin(5 * \pi/180) \\
&= 0.5 * 9.81 * 0.05 * 0.08716 \\
&= 0.24525 * 0.08716 \\
&= 0.0214 \text{ N.m}
\end{aligned}$$

B.9 Stability Factor

$$\begin{aligned}
\text{Factor} &= T_g / T_{\text{grav}} \\
&= 0.4571 / 0.0214
\end{aligned}$$

$$= 21.4 \sim 21 \quad (\text{STABLE -- criterion: factor} > 1.0)$$

B.10 Supply Voltage

$$\begin{aligned} k_e &= 1 / (KV * 2\pi/60) \\ &= 1 / (950 * 0.10472) \\ &= 1 / 99.48 \\ &= 0.01005 \text{ V.s/rad} \\ V &= k_e * \omega + I * R \\ &= 0.01005 * 314.16 + 10.0 * 0.12 \\ &= 3.157 + 1.200 \\ &= 4.36 \text{ V} \quad (\text{PWM duty} = 4.36/12 = 36.3\% \text{ at } 12 \text{ V supply}) \end{aligned}$$

B.11 Kinetic Energy

$$\begin{aligned} E_k &= (1/2) * I * \omega^2 \\ &= (1/2) * 0.002910 * (314.16)^2 \\ &= 0.001455 * 98677 \\ &= 143.6 \text{ J} \quad (= 0.040 \text{ Wh}) \end{aligned}$$

B.12 Bearing L10 Life

$$\begin{aligned} P_{\text{radial}} &= (m * g) / 2 = (0.5 * 9.81) / 2 = 2.45 \text{ N} \\ L_{10} &= (C/P)^3 * (10^6 / (60 * N)) \\ &= (3450/2.45)^3 * (10^6 / (60 * 3000)) \\ &= (1408.2)^3 * 5.5556 \\ &= 2.792 \times 10^9 * 5.5556 \\ &= 1.551 \times 10^{10} \text{ revolutions} \\ &= 1.551 \times 10^{10} / (3000 * 60) \\ &= 85,900 \text{ hours} \quad (\text{bearing life not a design constraint}) \end{aligned}$$

B.13 Thermal Analysis

$$\begin{aligned} P_{\text{copper}} &= I^2 * R = (3.0)^2 * 0.12 = 1.08 \text{ W} \\ A_{\text{casing}} &= \pi * D * L = \pi * 0.025 * 0.030 = 0.002356 \text{ m}^2 \\ \Delta T &= P_{\text{copper}} / (h * A) \\ &= 1.08 / (10 * 0.002356) \\ &= 1.08 / 0.02356 \\ &= 45.8 \text{ degC} \\ T_{\text{surface_max}} &= 30 + 45.8 = 75.8 \text{ degC} \quad (< \text{motor limit of } 100 \text{ degC winding}) \end{aligned}$$

B.14 Shaft Hoop Stress at Overspeed

$$\begin{aligned} \sigma_{\text{hoop}} &= (3+\nu)/8 * \rho * \omega^2 * r_o^2 \\ &= (3.3/8) * 7850 * (392.7)^2 * (0.1079)^2 \\ &= 0.4125 * 7850 * 154227 * 0.011642 \\ &= 0.4125 * 7850 * 1795.5 \\ &= 5.81 \text{ MPa} \end{aligned}$$

Safety factor = $400 / 5.81 = 68.8$ (very safe)

Appendix C: Project Gantt Chart

The project was conducted over two academic semesters (approximately 28 weeks). Major phases and their durations:

Phase	Weeks	Key Deliverables
Literature review	1-4	Comparative tables; method selections
Conceptual design	3-6	Subsystem definitions; BOM first draft
Detailed design + calcs	5-9	Full calculations; SOLIDWORKS drawings
Simulink simulation	8-12	PID tuning; stability simulation
Proteus circuit simulation	9-12	ESC model; circuit verification
Material procurement	8-11	All components ordered and received
Flywheel machining	11-14	CNC turning; balancing
Frame modification	12-14	Line-boring; welding; painting
Electronic assembly	13-16	Perf-board; soldering; testing
Firmware development	10-18	ESP32 PID; Kalman; Wi-Fi; serial

Server and dashboard	10-18	Node.js; PostgreSQL; React; WebSocket
Integration and tests 1-4	17-20	All subsystem tests completed
Extended stability test	20-21	Test 5 completed
Thesis writing	18-28	Parallel with testing and software phases

Appendix D: Firmware Overview and Key Code Structures

The firmware is written in C++ for the Arduino/ESP32 framework and compiled with PlatformIO. The main source files are:

File	Purpose
main.cpp	Entry point; setup(); loop(); task scheduling
imu.cpp / .h	MPU6050 I2C driver; raw data read functions
kalman.cpp / .h	Single-axis Kalman filter implementation
fft_rpm.cpp / .h	arduinoFFT wrapper; RPM estimation from accel FFT
pid.cpp / .h	Discrete-time PID with anti-windup and derivative filter
esc.cpp / .h	ESP32 LEDC PWM output; microsecond pulse generation
telemetry.cpp / .h	JSON serialisation of all sensor values
serial_cmd.cpp / .h	Command parser for UART and Wi-Fi TCP channels
wifi_client.cpp / .h	ESP32 Wi-Fi STA connect; TCP socket management

Key firmware constants and their values used during the endurance run:

Constant	Value
LOOP_PERIOD_MS	20 (50 Hz PID and IMU rate)
TELEMETRY_PERIOD_MS	100 (10 Hz telemetry transmission)
FFT_SAMPLE_COUNT	64
FFT_SAMPLE_RATE_HZ	500
ESC_MIN_US	1050
ESC_MAX_US	2000

PID_KP	0.85
PID_KI	2.40
PID_KD	0.015
SETPOINT_RPM	3000
KALMAN_Q	0.001 (process noise)
KALMAN_R	0.030 (measurement noise)

Appendix E: Data Sheet List

The following data sheets were consulted during component selection, circuit design, and bearing life calculation:

- ESP32 Technical Reference Manual v5.1 -- Espressif Systems, 2023
- ESP32 Datasheet v3.4 -- Espressif Systems, 2024
- MPU-6050 Product Specification Revision 3.4 -- InvenSense, 2013
- RS608 Deep Groove Ball Bearing -- RS Components, 2023
- IRFZ44N HEXFET Power MOSFET Datasheet -- International Rectifier
- IR2101 Half-Bridge Gate Driver Datasheet -- International Rectifier
- LM7805 Voltage Regulator Datasheet -- Texas Instruments, 2023
- AMS1117-3.3 LDO Regulator Datasheet -- Advanced Monolithic Systems
- 950 KV x2212 Outrunner BLDC Motor Specification -- Manufacturer spec sheet

Appendix F: Safety Assessment and Laboratory Setup

A formal risk assessment was completed and approved by the supervisor before any high-speed testing began. The assessment identified five credible hazard scenarios (listed in Table 22, Chapter 5) and defined mitigations for each.

Physical safety measures implemented in the laboratory:

1. Polycarbonate safety screen:

6 mm polycarbonate sheet on an aluminium extrusion frame, enclosing three sides of the gyroscope. The fourth side faced the laboratory wall. The screen was rated to contain a fragment ejected at the flywheel rim speed (approximately 90 m/s at 3000 RPM).

2. Operator positions during testing:

During spin-up and spin-down, operators stood to the side of the gyroscope, not directly in front of or behind the spin axis, to minimise exposure to the most likely fragment ejection paths.

3. Fused power supply:

A 3 A fast-blow fuse was installed in the motor supply positive lead. This was verified to blow correctly during a simulated short-circuit test before the endurance run.

4. Emergency stop access:

The dashboard emergency stop button was visible and accessible from the monitoring position at all times. The physical power supply disconnect switch was also accessible without approaching the safety screen.

Appendix G: Source Code and File Repository

All project files are archived in the following locations:

Resource	Location
Source code	https://github.com/01chabotamwenda-commits/gyro.git
MATLAB models	CBU Google Drive -- shared with supervisor
Proteus files	CBU Google Drive -- shared with supervisor
Test data (SQL)	PostgreSQL backup -- project storage drive
SOLIDWORKS files	CBU Google Drive -- shared with supervisor
This report	CBU School of Engineering submission system